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1. Introduction:

1.1 Introduction to the Topic

Administrative tasks like maintaining student attendance, handling grades, processing funds, and keeping records are frequently done by hand in traditional educational institutions. These procedures usually use outdated software or paper-based methods, which results in inefficiencies, human error, and a delay in information exchange (theschoolmanagement, 2024). There is little opportunity to enhance the overall educational experience because teachers and administrators spend a lot of time on repetitive tasks.

The School Management System (SMS), on the other hand, provides an advanced solution that organizes and automates these essential tasks, greatly improving the accuracy and efficiency of school operations. SMS removes the need for manual data entry and paper-based records by digitizing procedures including student enrollment, attendance tracking, grade management, and fee collecting. Additionally, the system guarantees that parents, teachers, and kids have real-time access to critical information, which promotes improved communication and prompt decision-making. The SMS additionally shields private information from unwanted access thanks to its integrated security features. The School Management System is an essential tool for modern schools since, in contrast to old approaches, it improves overall management of educational institutions, minimizes errors, and streamlines administrative operations.

School Information Management System Market - Growth Rate by Region

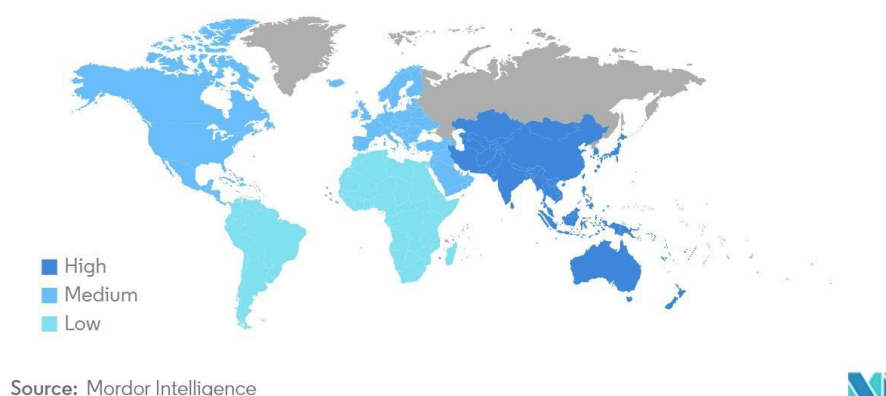


Figure 1: Increase in schooling management in system (MordorIntelligence, 2024)

1.2 Current Scenario:

In the era of digitalization, educational institutions are under increasing pressure to improve student engagement and optimize their administrative procedures. Traditional systems frequently use manual processes, which cause delays, mistakes, and inefficiencies. A study published by the U.S. Department of Education's National Center for Education Statistics, as of 2022, 75% of public schools reported using cloud-based systems for at least some of their administrative functions (Verified Market Research, 2024).

1.3 Problem Statement :

In Nepal, however, many educational institutions still face significant challenges due to a lack of streamlined platforms for managing various educational and administrative tasks. As a result, administrators and students experience a decentralized approach to communication and data management. These inefficiencies can hinder academic achievement, lower student satisfaction, and limit the potential for institutional growth and development. (edigitalnepal, 2024)

The need for modernization is becoming more and more obvious, even if Nepal's educational system is digitizing more slowly than other developed countries. Many institutions encounter obstacles including inadequate funding and shortages in technology expertise. However, the advantages of switching to a digital system enhancing communication, streamlining administrative procedures, and improving learning outcomes are becoming increasingly clear.

1.4 The project as a solution:

This is where this project is useful. By providing a comprehensive, cloud-based solution that improves administrative tasks and student participation, the proposed educational system seeks to eliminate the current inefficiencies. The system will have several features that are intended to automate and streamline important procedures in educational establishments, facilitating communication between staff, students, and administrators. These characteristics, which provide flexibility and scalability for educational institutions of various sizes, are specifically designed to meet the demands of Nepal's educational environment.

1.5 Aims and Objectives

1.5.1 Aim:

Enhancing Traditional schooling system: This project's main goal is to create a School Management System that improves student involvement through constant updates and notifications while reducing administrative operations. The system will use modern technology to automate essential processes including student enrollment, attendance tracking, grade management, and fee collecting.

1.5.2 Objectives

1. **Affordable Access to Technology:** By end of April 2024, design and implement a cost-effective educational management system that aligns with the budget constraints of 80% of educational institutions in Nepal, ensuring its affordability and sustainability.
2. **Improvement of Administrative Efficiency:** Reduce the amount of paperbased administrative work by 50% within the first six months of deployment by automating procedures including event registration, fee collecting, student enrollment, and attendance tracking.
3. **Commitment to the law and regulations:** Verify that the system satisfies all local laws regarding security, privacy, and education under Nepal's Personal Data Protection Act.
4. **Promoting Student Engagement and Learning:** E By implementing features like event registration, exam management, and real-time notifications of academic activities, increase student involvement by 40% in the first academic year.
5. **Automated Attendance Recording (Future Work):** In the coming Future integrate a fingerprint-based attendance system that automates attendance tracking, aiming to eliminate manual record-keeping and reduce errors by 90%.
6. **Administrative Tools:** Launch a complete admin page with CMS features, enabling administrators to efficiently manage notifications, schedule events, and create and update content, reducing administrative workload by 30%.

2. Background:

2.1 About the End User/ Client:

The primary end users of the proposed School Management System (SMS) include the following stakeholders within educational institutions:

❖ School Administrators

Leaders in a school have the mandate of overseeing the general running of the school through overseeing issues such as students' attendance, fees, grades, and records. The SMS will assist the administrators by eliminating bureaucracy that comes with human errors together with increasing efficiency and establishing a control point for all the administrative activities. They will then be able to spend more time on chronic strategic stakeholder planning and decision-making for institutional more time.

❖ Teachers

This is a primary secret of teachers in that they are often working on a number of fronts in the classroom, marking assignments, in having to ensure that students attend their classes as expected. The SMS will automate these processes by enabling digital documentation of mechanical work; providing instant student information; and enable grading. This will enable teachers to spend most of their time offering quality education and quality time to students.

❖ Students

Any learning institution revolves around learners and that is why the SMS puts the focus on students. Amid free access to grades and attendance, as well as school announcements and other important notices, students can be aware of their progress and engage in their schooling process. The SMS will also assist the students to facebook with the teachers and to get access to other resources easil

System in Nepalese Educational Institutions

The target clients of the SMS are mainly educational institutions of Nepal: private, relatively small schools and large state schools. Such institutions struggle with the issues, such as separated communication, the absence of effective administrative hierarchy, and the restricted usage of progressive technology. To overcome these challenges, the SMS is designed to meet these needs sustainably at a low cost and integrates with the context of Nepalese education systems.

2.2 Similar Project:

In this section, discuss of some of the school management systems and educational technology were looked into that have been adopted to improve administrative hindrances and students' interaction. These projects are similar of the proposed e-School system for a community college, It is essential for the educational management to apply information systems in order to build efficient relationship between schools; the discussed below projects are among the best examples of the successful implementation of information systems in educational environment.

2.2.1. Fedena Community Edition

Fedena is an open-source educational management software through which many schools can be operated easily through a number of services like student management, grade reports, absent records, and communication solutions. Fedena is the tool which helps in automating all the working process and grants the students, teachers, parents immediate data access.

❖ Key Features:

- Student and staff management.
- School calendar management
- Timetable management
- Online fee management

❖ Relevance:

Some of the features of Fedena like attendance management, were referenced as it suited the objectives of the system.

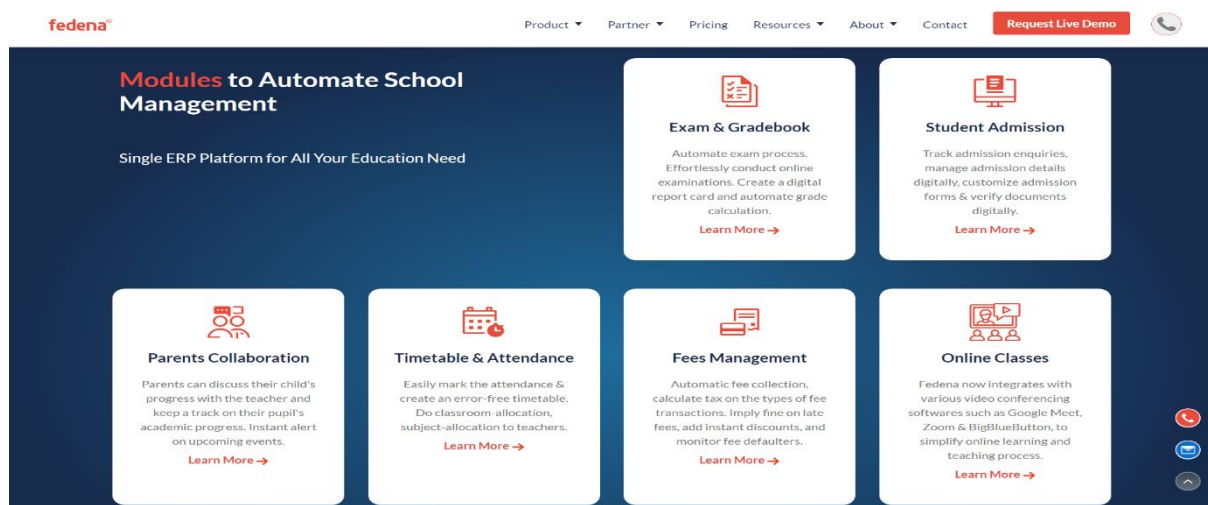


Figure 2: Fedena Schooling system

2.2.2 OpenSIS

Overview:

OpenSIS is a open-source student information system which is suitable for K-12 schools and colleges and universities. For tangible benefits, it enables an institution to keep track of records such as student data, performance, attendance, and it helps in scheduling). OpenSIS is a versatile product capable of easily scaling to meet the requirements of an institution.

Key Features:

- Student record management
- Generation of gradebook and report cards
- Attendance tracking

Relevance:

These are features on how OpenSIS can handle large amount of data, generate automatic reports, and attendance, are all closely related to the need for the system.

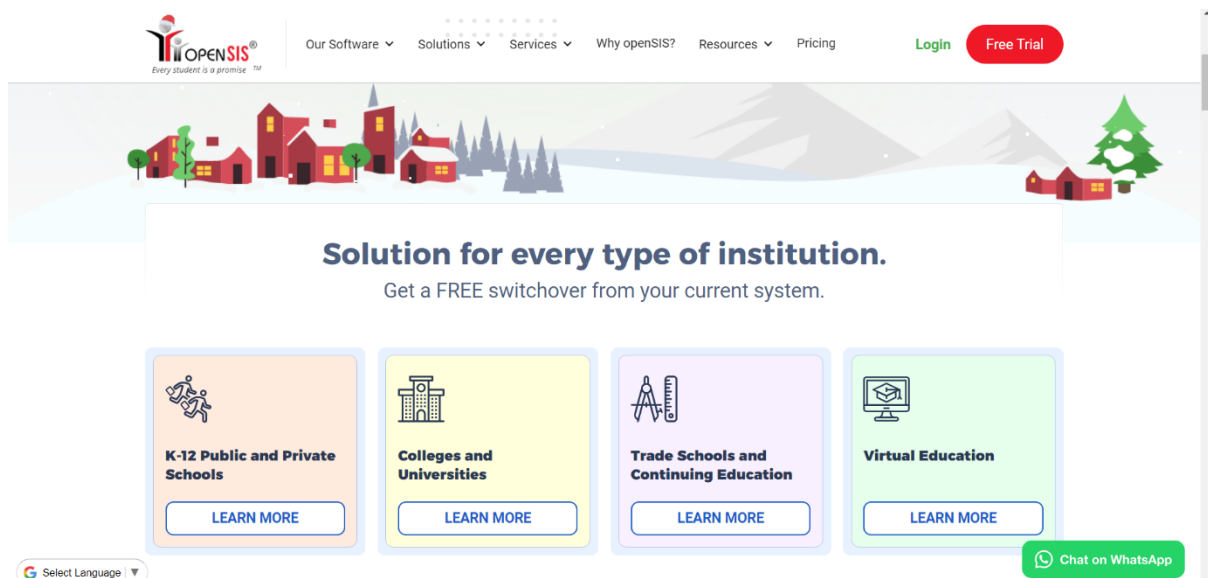


Figure 3: OpenSIS school management Software

2.2.3. Moodle

Overview:

Moodle is a Cloud based software solution providing tools for School Administration, this, system covers three major functions namely; attendance, grading, and communication functions and reporting. Moodle's main functions are also clear and accessible, its concept built around ease of use for school staff, as well as parents.

Key Features:

- Fee management
- Schedule management
- Communication features for parents, staffs and students

Relevance:

As the system heavily relies on administrative side the software greatly helped by giving ideas as to how the system flow works in admin sided pages

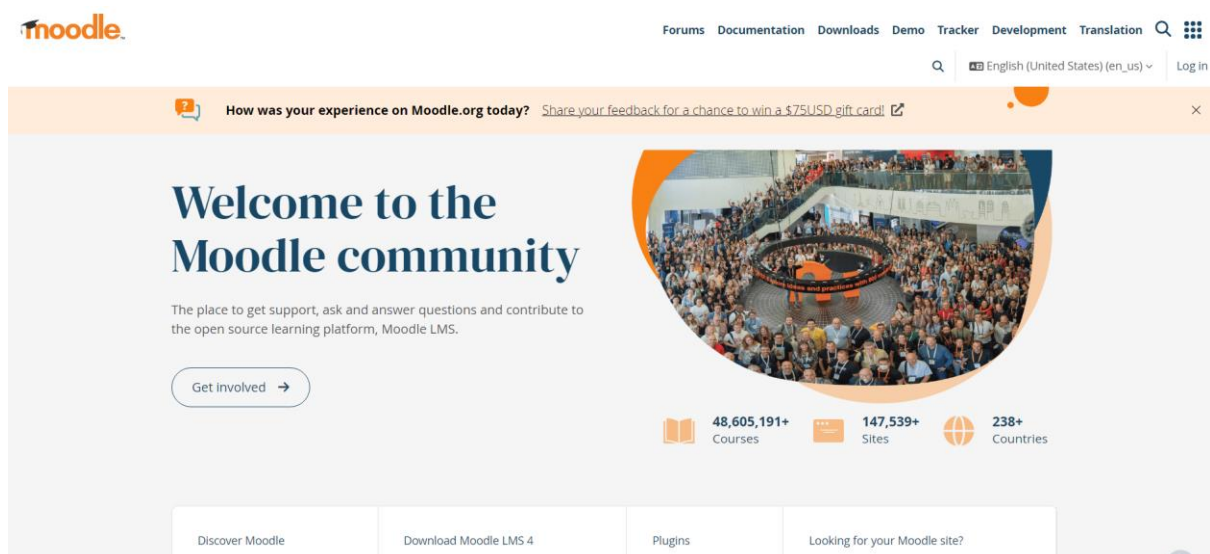


Figure 4:Moodle Software

2.3 Comparison Of the project:

Feature / Criteria	OpenSIS	Moodle	Fedena	Proposed e-School System
Primary Focus	Student Information Management (SIS)	Learning Management System (LMS)	School Management System (SMS)	Administrative Focused
Student Management	Yes	Yes	Yes	Yes
Attendance Management	Yes	Yes (via plugins)	Yes	Yes
Grade Management	Yes	Yes (Gradebook and report plugins)	Yes	YES
Fee Management	Yes	No (requires plugins)	Yes	Yes
Communication	Yes (via portal)	Yes (via plugins)	Yes	NO
Events	NO	YES (requires plugins)	NO	YES
Examination	NO	NO	YES (requires plugins)	YES
Timetable	Yes	Yes (via plugins)	Yes	Yes
Reporting and Analytics	Yes (customizable reports)	Yes (via plugins)	Yes (detailed reports)	Yes (real-time data and reports)

3. Methodologies:

The systematic approach or frameworks used to plan, design, develop, test, and maintain websites and online applications are referred to as methodologies in web development. They offer a methodical approach to project management, guarantee effective teamwork, and produce high-quality outcomes on schedule. Well-known approaches like Agile, Scrum, Kanban and Waterfall provide detailed instructions for allocating responsibilities, setting priorities for developments, and managing modifications. Methodologies are crucial in web development because they help control complexity, enhance team communication, guarantee consistency, and lower the possibility of mistakes, all of which contribute to the creation of more successful, efficient, and user-friendly online solutions.

3.1 Considered Methodology:

3.1.1 Feature-Driven Development (FDD):

Feature-Driven Development (FDD) was also considered for this project due to its focus on building software through feature-specific tasks, making it well-suited for modular components like user authentication, event registration, and notifications. This methodology provides clear deliverables and iterative progress, which aligns with the project's need for structured development. However, it was not selected because it is more suitable for larger teams and complex projects. Kanban, with its simplicity, flexibility, and visual task tracking, was a better fit for this smaller-scale project, ensuring efficient task management and adaptability.

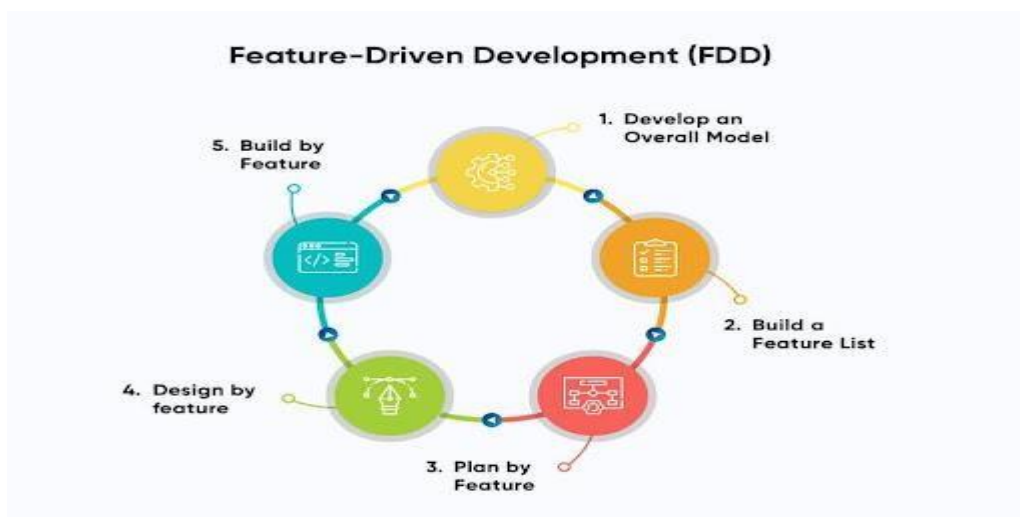


Figure 5: Development process for FDD model (Gorapid Labs, 20024)

3.1.2 Agile Scrum:

Scrum was taken into consideration because of its systematic approach to continuous improvement using sprints, which enables regular feedback and systematic progress for features like user authentication and event management. But because of its formal procedures and role requirements, which were too complicated for a small team, it was not selected. For this project, kanban's continuous process and simplicity were thought to be more suitable.

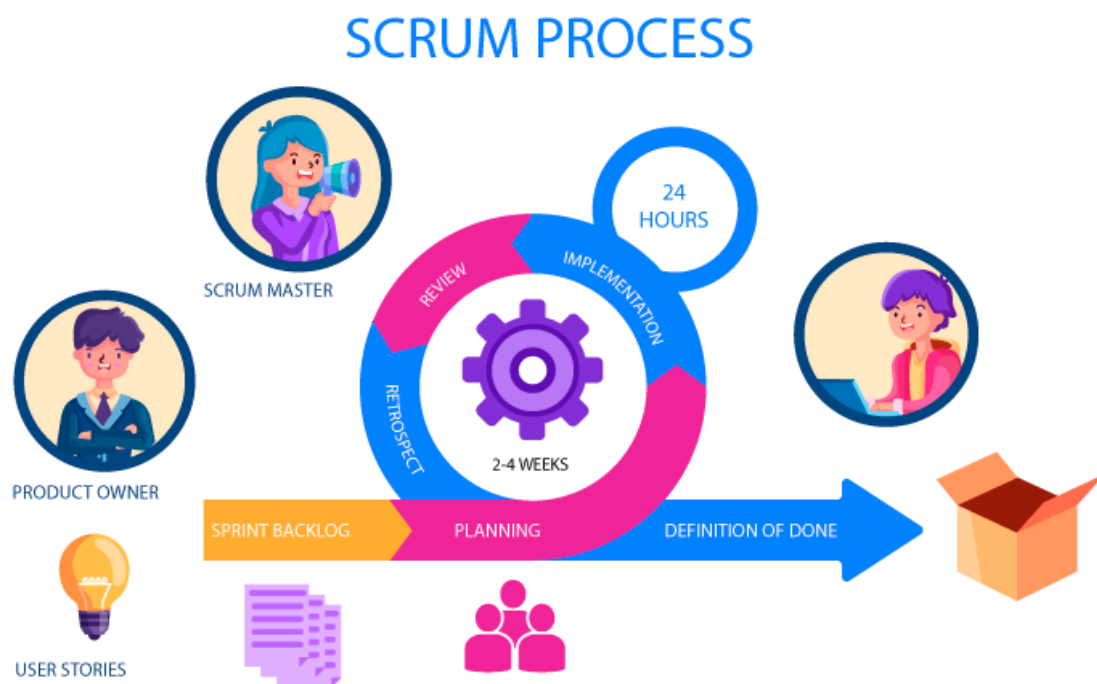


Figure 6: Process involved during scrum (Bhaskar S, 2024)

3.2 Selected Methodology:

Among various methodologies, Kanban from Agile was selected as Kanban is a methodology from Agile that helps organize work efficiently by visualizing tasks and identifying problems. It uses a visual board with task cards to track progress, reduce waste, and improve workflow. This structured approach ensures timely completion of tasks. Ultimately, Kanban boosts productivity and quality.

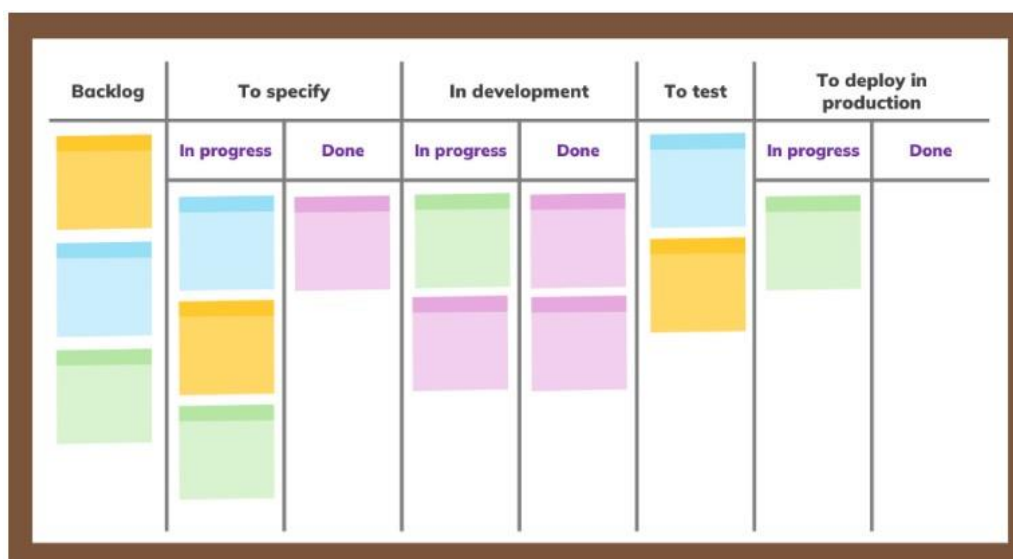


Figure 7: Kanban Board for Software development (Simon Buehring, 2024)

Reason for choosing Kanban for the project:

- Kanban in this project helped to quickly identify which tasks, such as student enrollment and attendance tracking, were in progress or completed. This ensured that project milestones were easily monitored
- The nature of management software is very flexible and requires frequent changes and updates and kanban provided that flexibility by smoothly adjusting priorities without disturbing the progress of other ongoing tasks
- Kanban helped the project by focusing on important tasks like grade management and fee collection, ensuring that no time is wasted on unnecessary steps.
- Any task that was delayed became visible on the Kanban board. This helped to quickly address and resolve issues, ensuring the project was completed on time.
- Due to the nature of Kanban, we are focusing on making small, gradual improvements in the project to avoid big changes to avoid disrupting the entire system. A through testing of system, Refined UI was achieved

3.3 Work Break Down Structure:

A Work Breakdown Structure (WBS) divides a project into smaller, manageable tasks in a structured way. It provides a clear hierarchy of activities, helping teams understand the scope of work. WBS ensures every part of the project is accounted for and organized effectively.

How WBS Helped in My Project:

1. Simplified task tracking and ensured smooth progress by clearly defining and organizing each phase of the project.
2. Helped identify dependencies and allocate resources efficiently for timely completion.

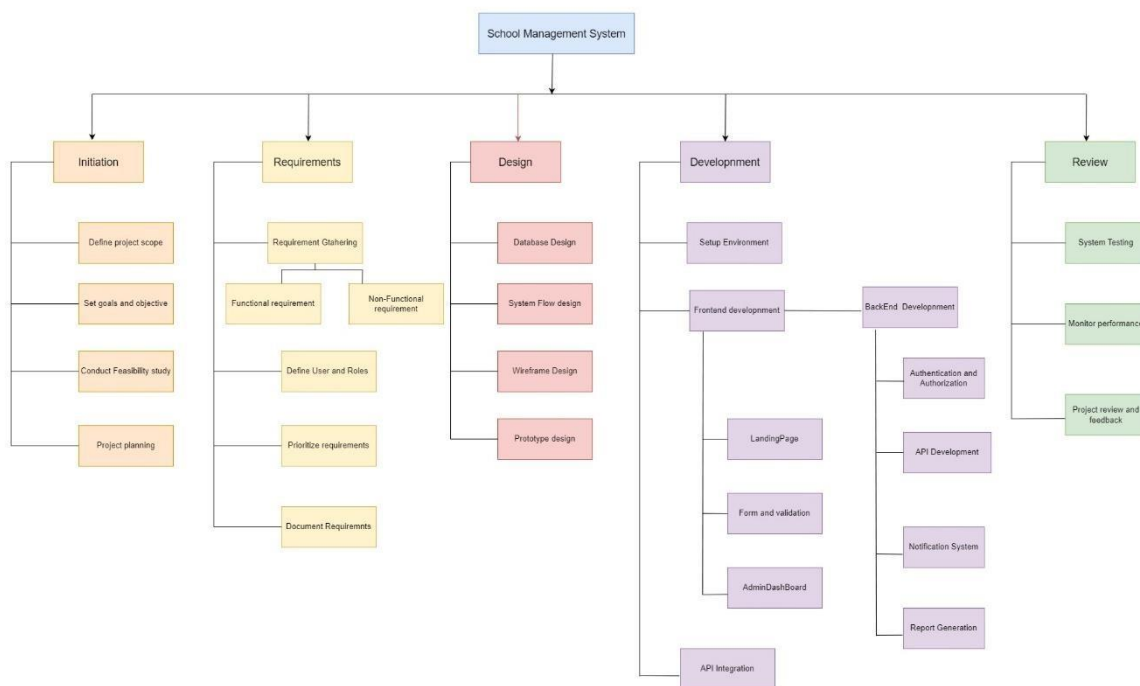


Figure 8: WBS image for the system

3.4 Gantt chart:

The chart below illustrates the work breakdown structure organized by time. As outlined in the milestones section, several key milestones are established throughout the project's duration. The project is expected to take approximately six months to complete, beginning with the initial research and planning phase and concluding with the final project submission. The timeline covers all critical stages, from securing the client and defining requirements to development, testing, and final reporting.

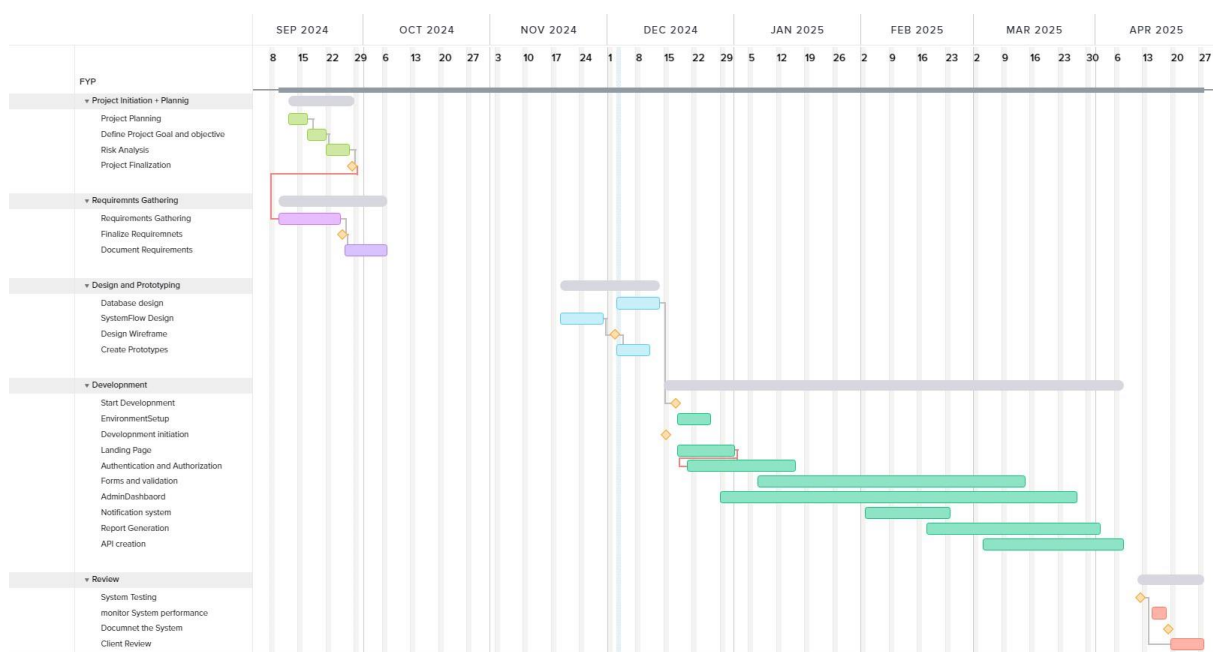


Figure 9: Gantt Chart System

4. Analysis Review:

4.1 Progress Summary:

The project of the development of the school management system website was initiated in order to provide users with an efficient platform to manage the work of schools and to function as a valuable tool effectively, and to achieve both the aims, the project management method called Kanban was used. The project began in early November and the major goal set during the project's initiation was to create the system that handles user registration and login. Significant advancements have been achieved in the development of the project, with several key milestones completed as outlined below:

- Completed Tasks Login and Registration System: The independent login and the registration system have been established with great efficiency.
- To enhance security, safety and privacy the systems have incorporated One-Time Password (OTP) verification.
- One additional, the "Remember Me" option has also been included to increase convenience for a user.
- Home Page Development: The home page also has been created and put into practice that provides a clear and effective structure of website to enhance the visual aesthetics for users.

Current Progress:

- Event Page Development: Work on the Event Page is underway at the moment. The functionalities to be created pertain to user interfaces' design as well as their implementation concerning event management. Some of the features being worked on include creation of events, editing, and display of events.

Next Steps

- Finalize the Event Page Development:
- Complete the implementation of event creation, editing, and viewing functionalities.
- Notification System Development:

4.2 Progress Table:

SN	Task	Status	Progress
1.	Project Planning	Completed	100%
2.	Define Project Goal and Objective	Completed	100%
3.	Risk Analysis	Completed	100%
4.	Project Finalization	Completed	100%
5.	End User and Target Finalization	Completed	100%
6.	Requirements Gathering	Completed	100%
7.	Client Specification Finalization	Completed	100%
8.	Design Schema and tables	Completed	100%
9.	Create Workflow Diagrams	Completed	40%
10.	Design Wireframe	Ongoing	40%
11.	Create prototypes	Ongoing	40%
12.	Start Development	Completed	100%
13.	Environment setup	Completed	100%
14.	Authentication And authorization	Completed	100%
15.	OTP verification	Completed	100%
16.	Homepage	Completed	100%
17.	Forms and validation	-	0%
18.	Events and registration	-	0%
19.	Fees and Payment Portal	-	0%
20.	Notification	-	0%
21.	Admin dashboard	-	0%
22.	Report and analysis	-	0%
23.	RESTful API'S	-	0%
24.	Integrate API	-	0%
25.	System Testing	-	0%
26.	Monitor Performance	-	0%
27.	Document the system	-	0%
28.	Client Review	-	0%

4.3 System Architecture:

The System Architecture Design focuses on how the components of the Blood Donation System should be deployed in order to meet the intended functionality. Basically, it explains the integration of the client-server and data layers, based on the Model-View-Template (MVT) structure that determines Django-based applications.

❖ Models (Data Layer)

Purpose: Represents the database schema and handles all interactions with the database.

- Each model corresponds to a database table.
- Django provides an ORM (Object-Relational Mapping) for easy querying.

❖ Views (Business Logic Layer)

Purpose: Acts as a bridge between models and templates, processing HTTP requests and returning HTTP responses.

- Contains the business logic.
- Processes data from models and sends it to templates for rendering.
- Can return JSON responses in APIs (via Django REST Framework).

❖ Templates (Presentation Layer)

Purpose: Handles the presentation logic and renders data into HTML, CSS, and JavaScript for the frontend.

- Use Django's template language to dynamically generate content.
- ❖ Interact with data passed from the views.

Control Flow Of MVT

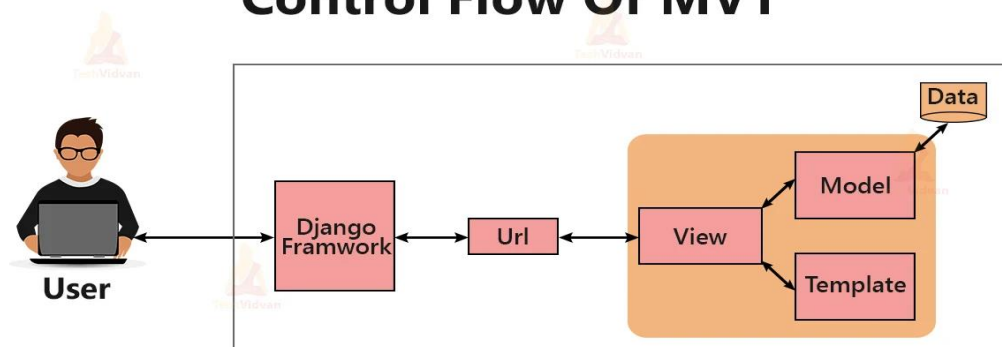


Figure 10: MVT flow in Django (Pain, n.d.)

4.4 ERD:

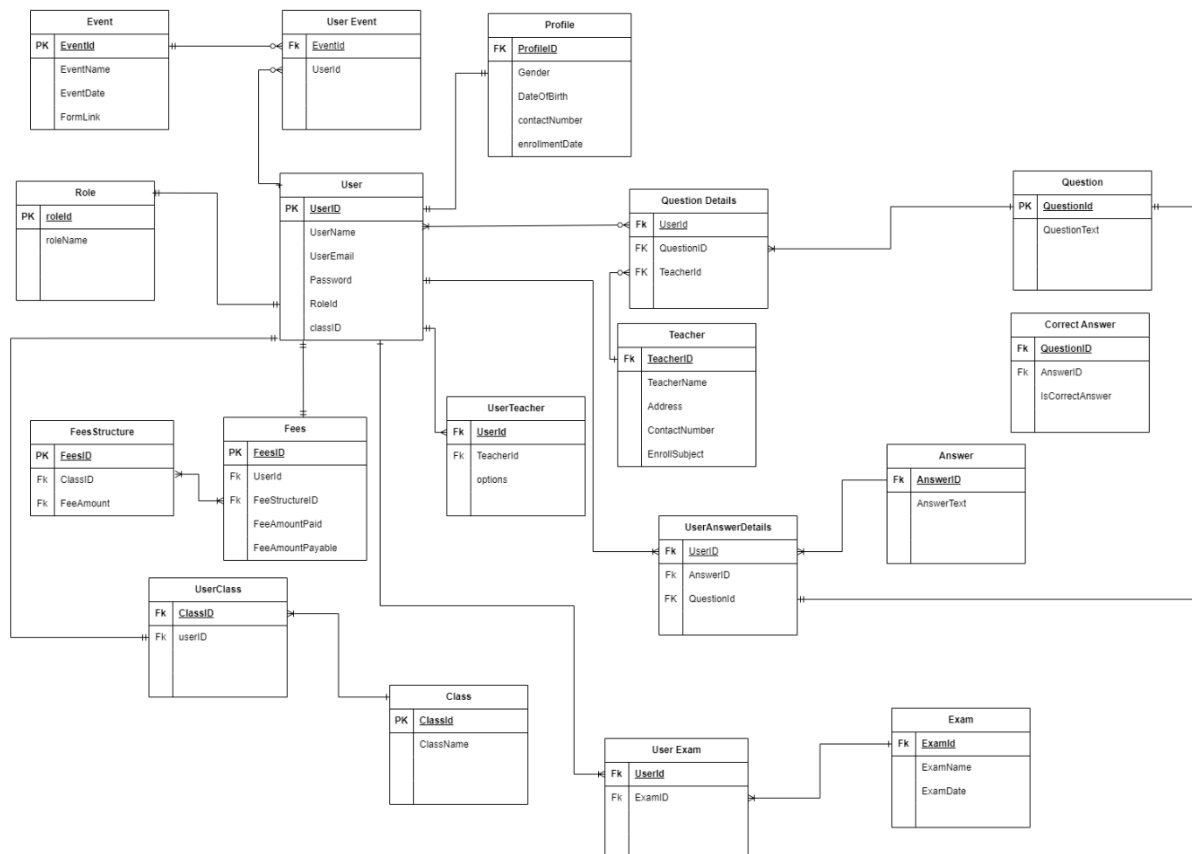


Figure 11:ERD for the system

List OF Entities Used:

- **USER**: Stores data of all user with in the system\
- **EXAM**: Stores all the data of exam details
- **CLASS**: Holds information about and class
- **FEE**: Stores users fee data
- **FEESTRUCTURE**: Holds fee structure of every different classes
- **QUETSION**: Stores question answer to be used in an exam
- **EVENTS**: Has list of all upcoming event
- **EVENTPASRTICIPTAION**: gets list of all student who participated in a event

4.5 Use Case Diagrams:

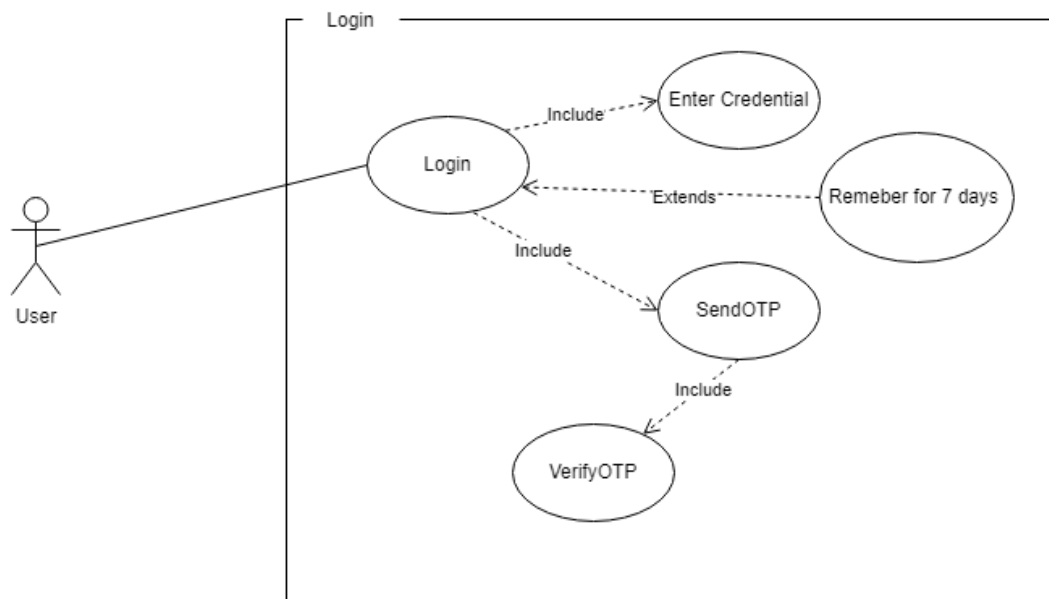


Figure 12: Use case for Login Process

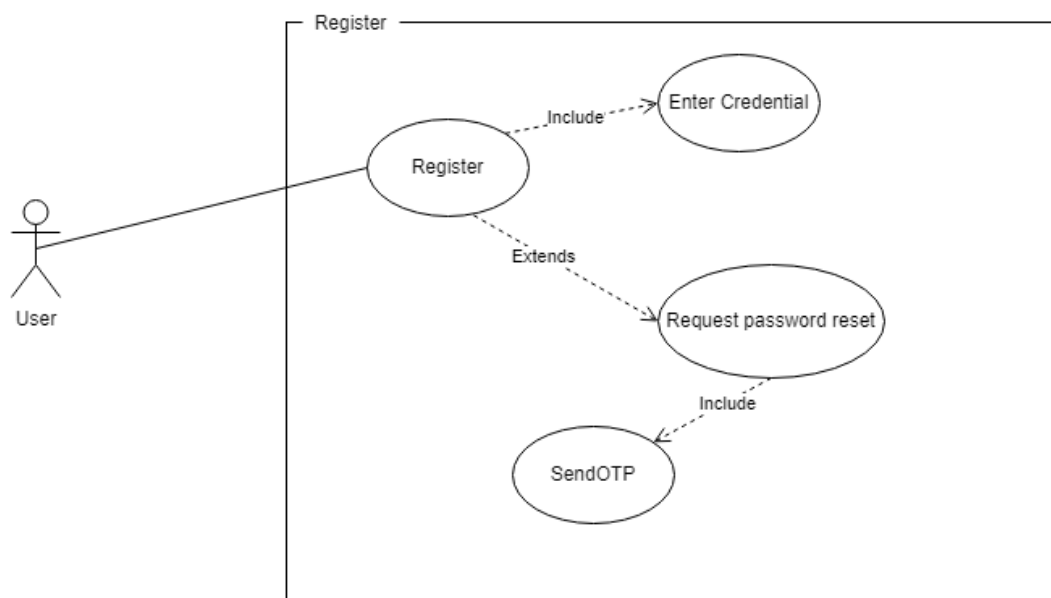


Figure 13: UseCase For register

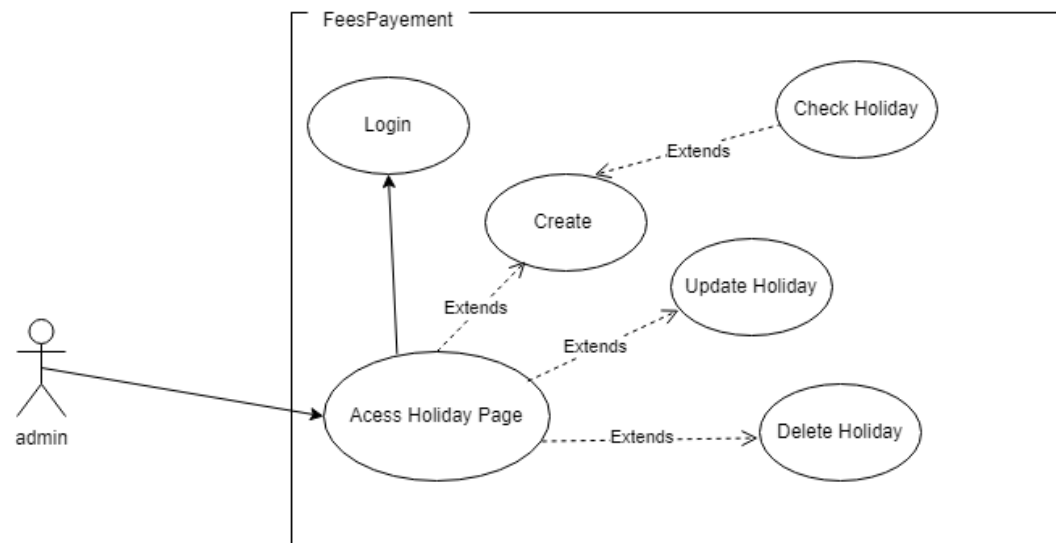


Figure 14: Use Case for holiday Creation

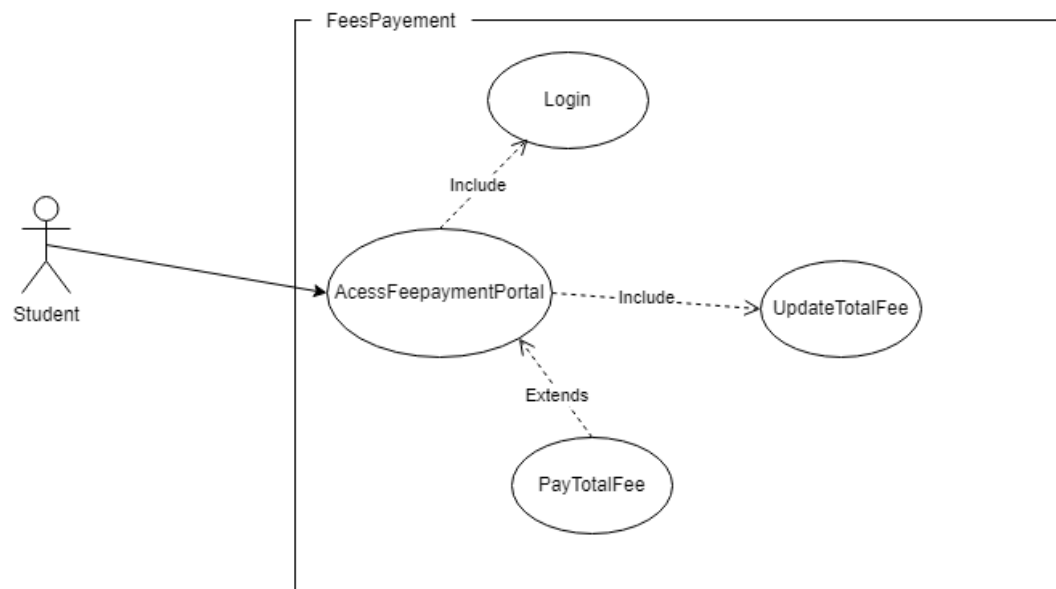


Figure 15: UseCase For Fee Payment

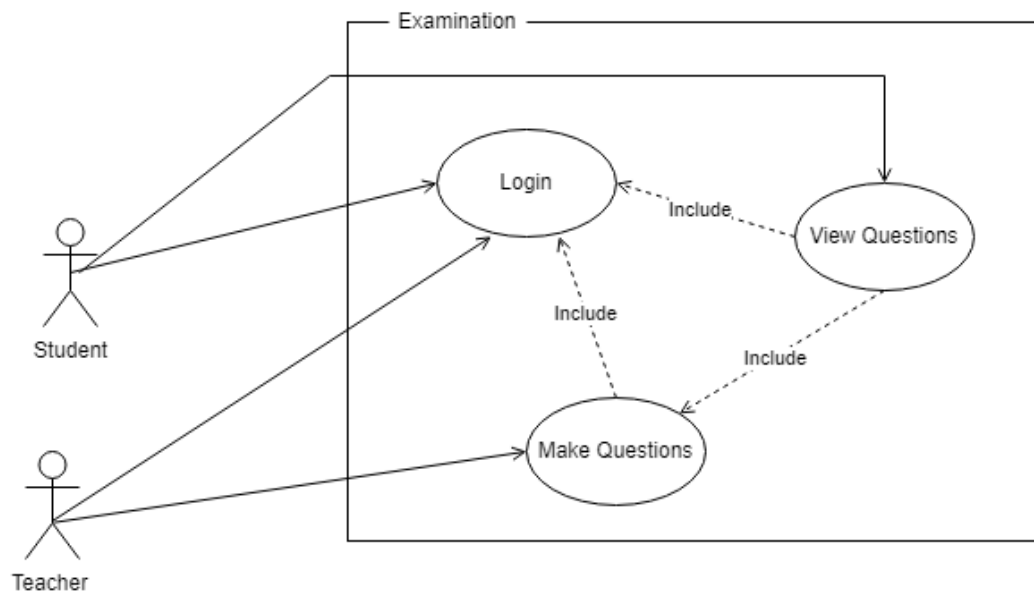


Figure 16: Use For Examinations

4.6 Sequence Diagram:

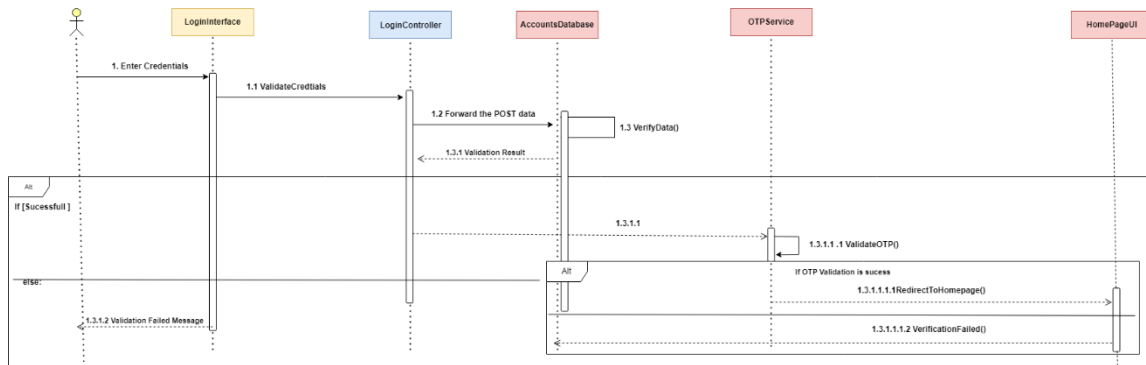


Figure 17: Sequence for login

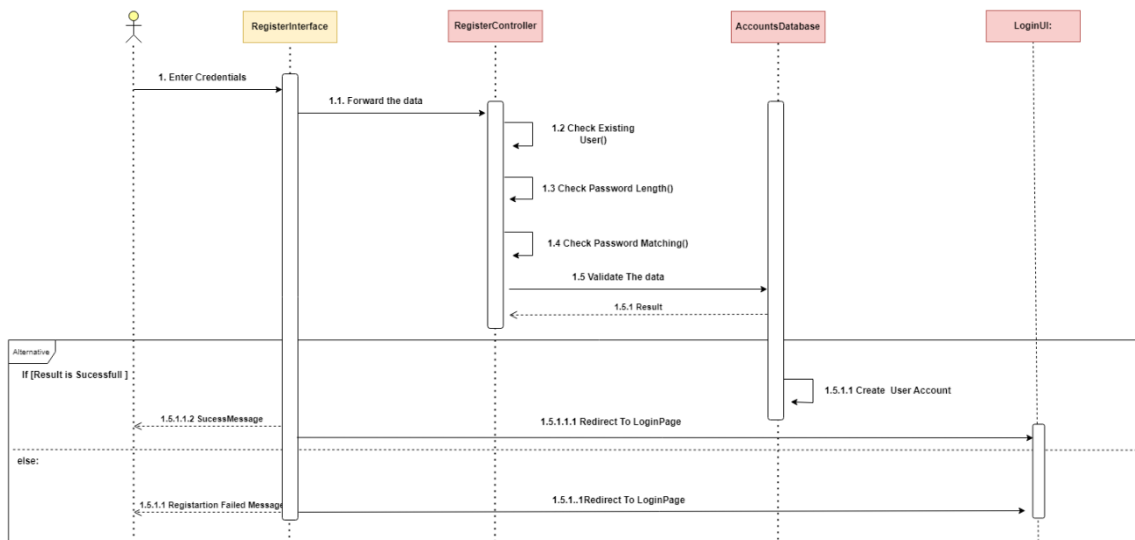


Figure 18: Sequence for Register

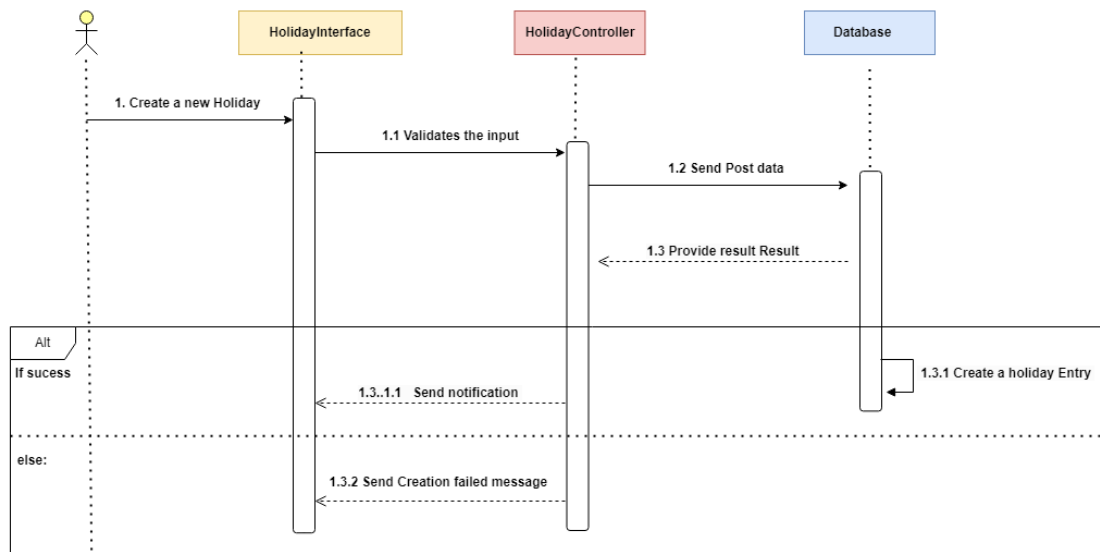


Figure 19:Sequence For holiday

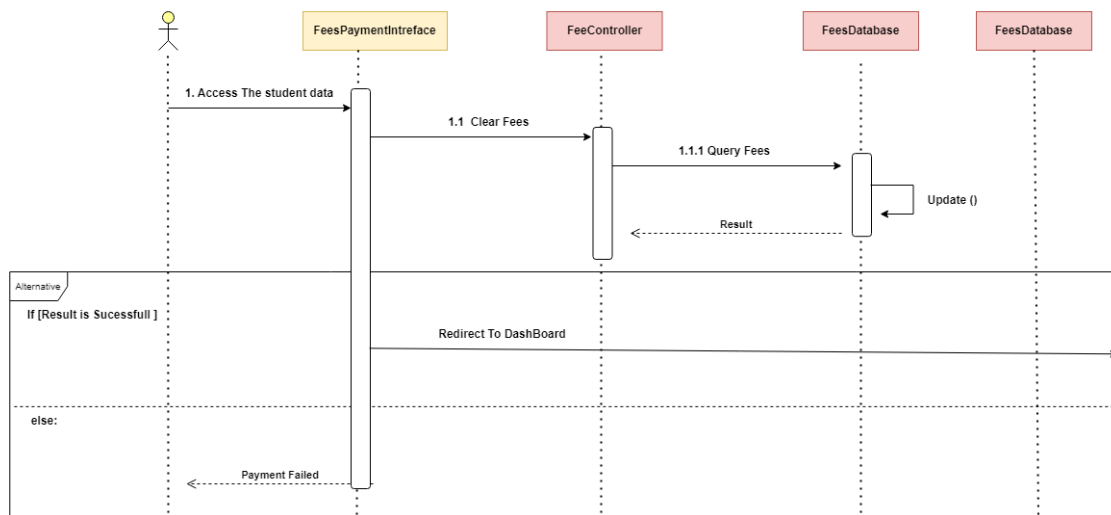


Figure 20:Sequence for Fee Payment

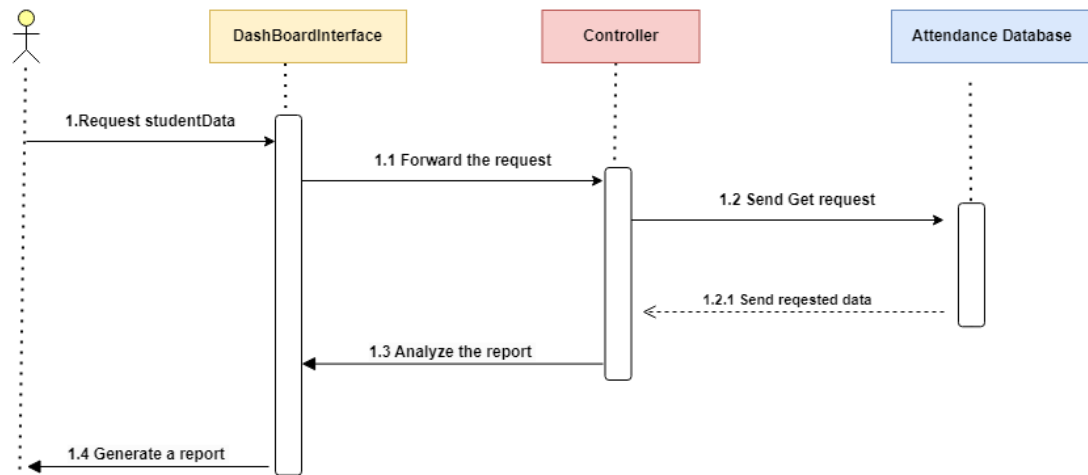


Figure 21:Sequence for Report generation

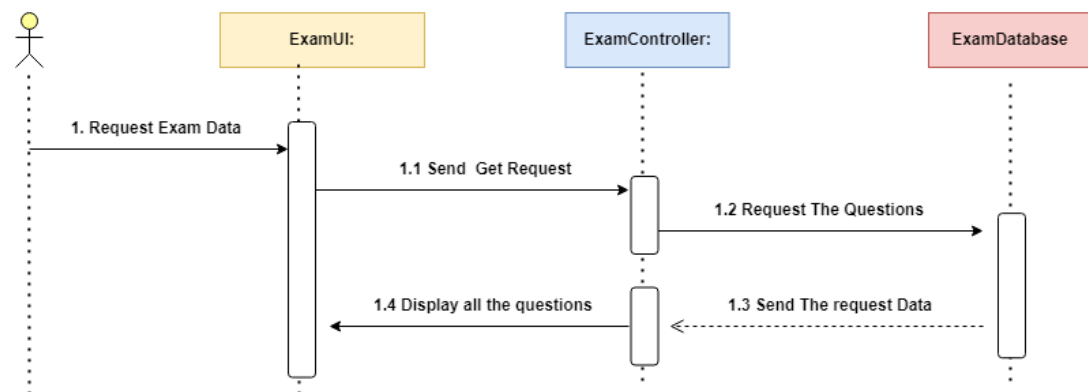


Figure 22:Sequence for Examination UI

4.7 Activity Diagram:

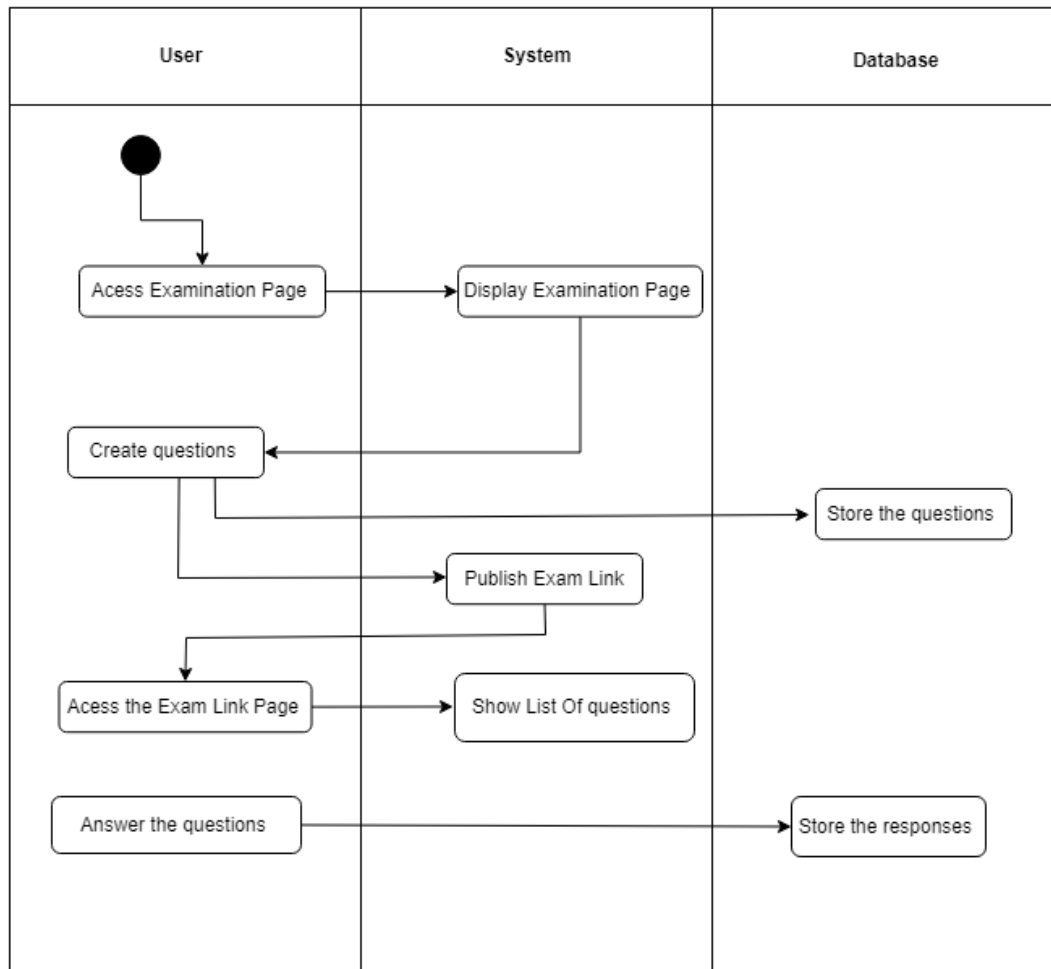


Figure 23:Activity diagram for Examination

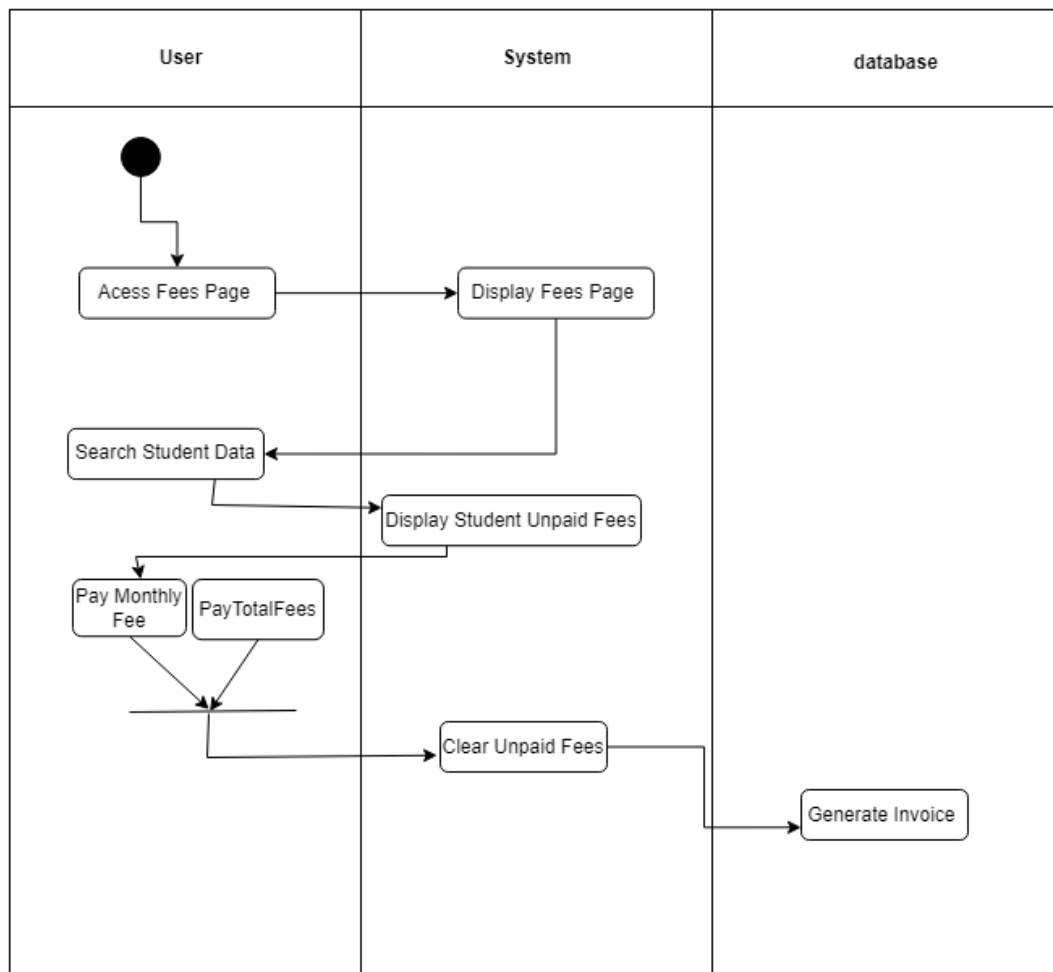


Figure 24:Activity diagram for make payment

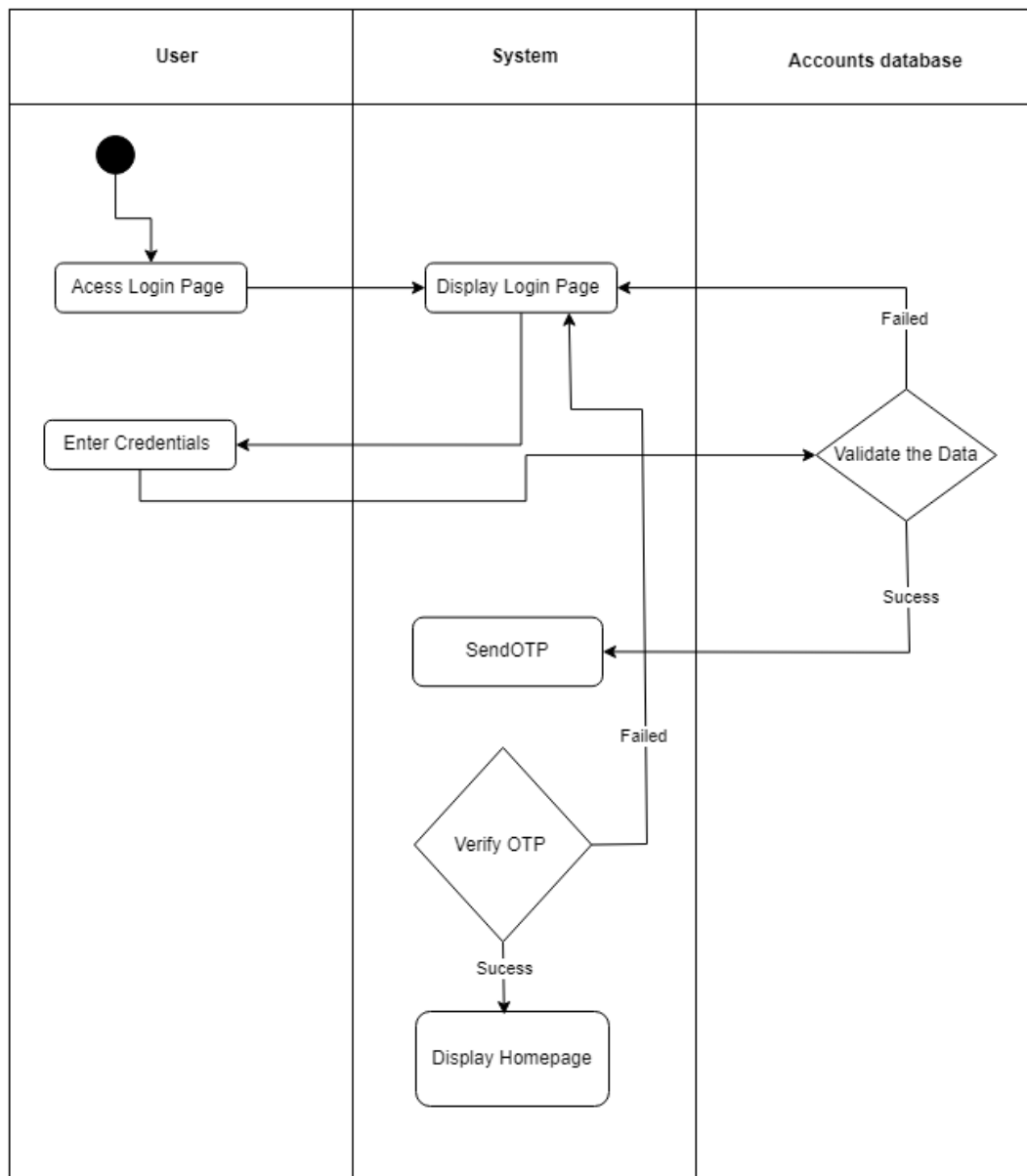


Figure 25:Activity for Login Page

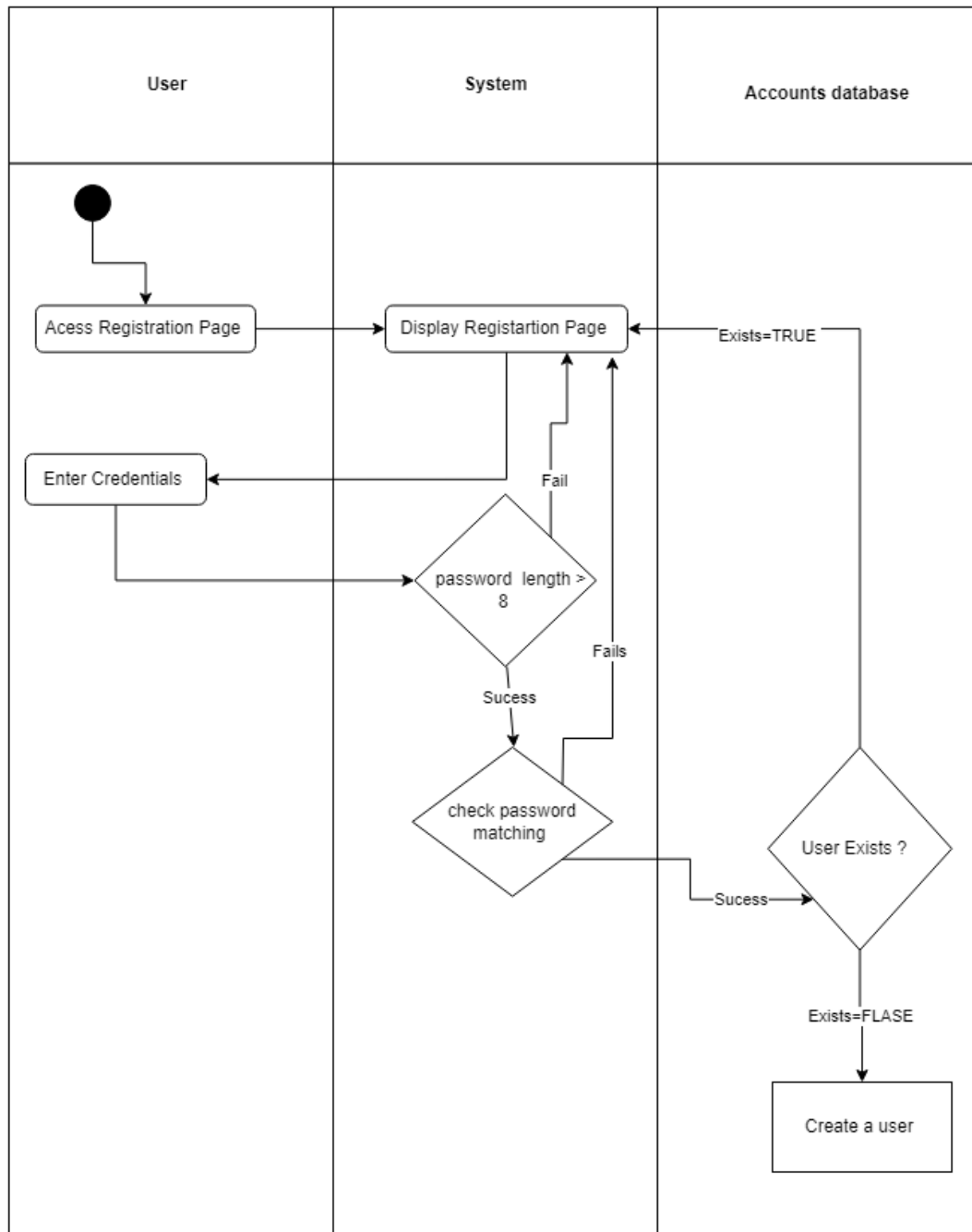


Figure 26:Activity UML for register

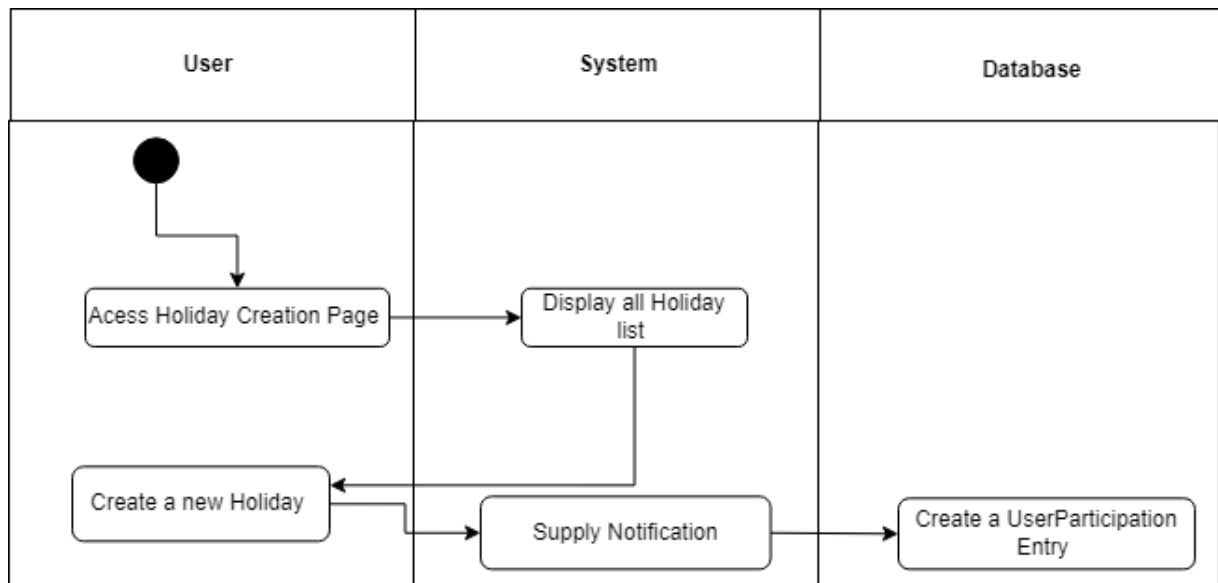


Figure 27:Activity for Holiday

4.8 Collaboration Diagram:

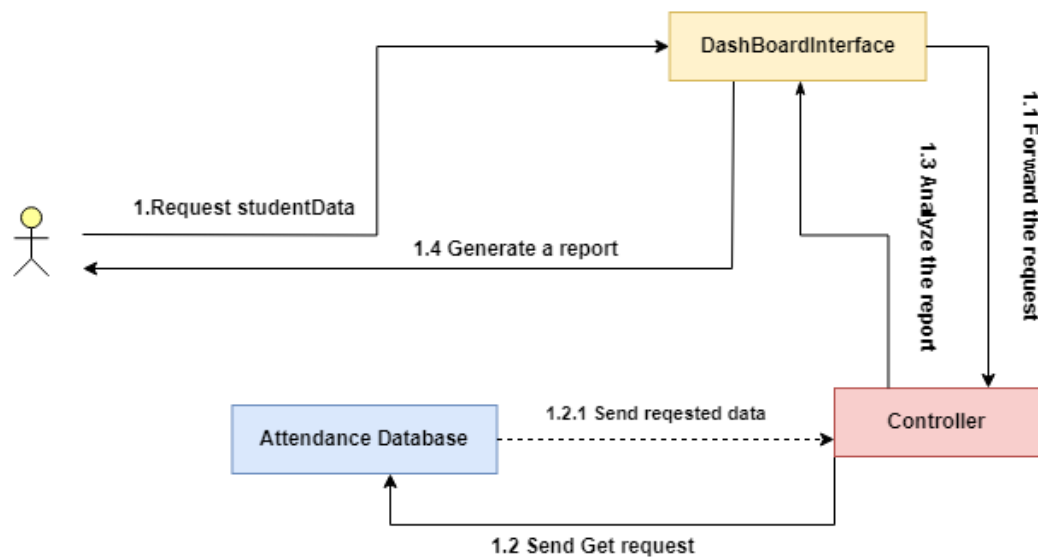


Figure 28:Collaboration for Report Generation

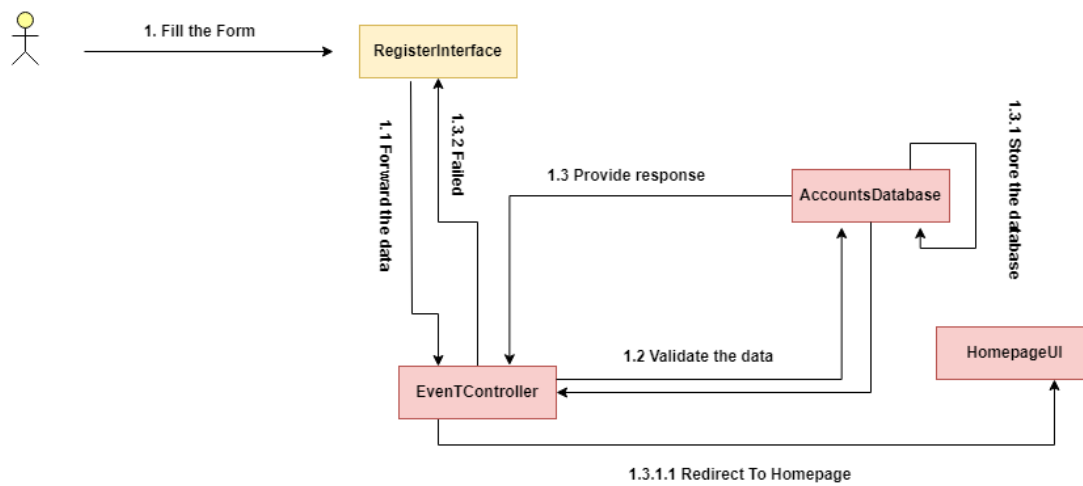


Figure 29:Collaboration for Event

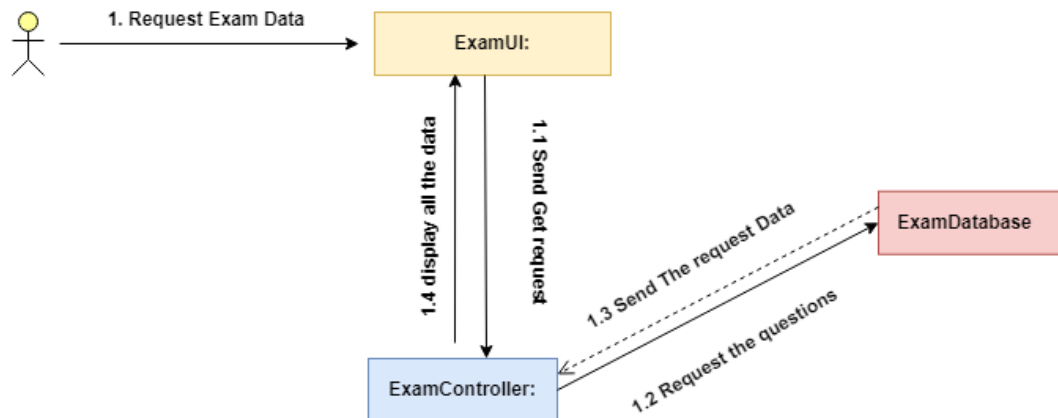


Figure 30:Collaboration for Examination

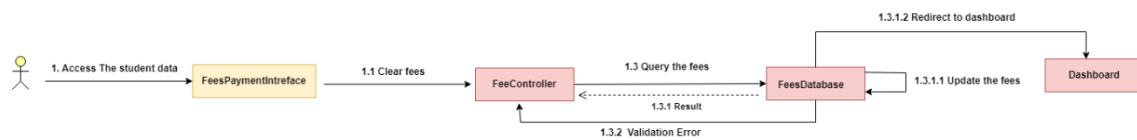


Figure 31:Collaboration for FeesPayment

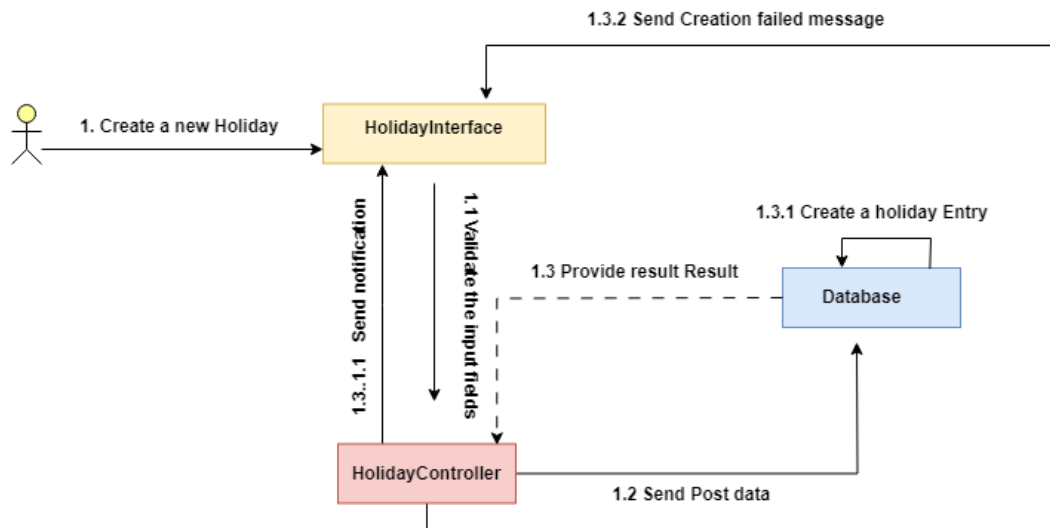


Figure 32:Collaboration for Holiday Creation

S

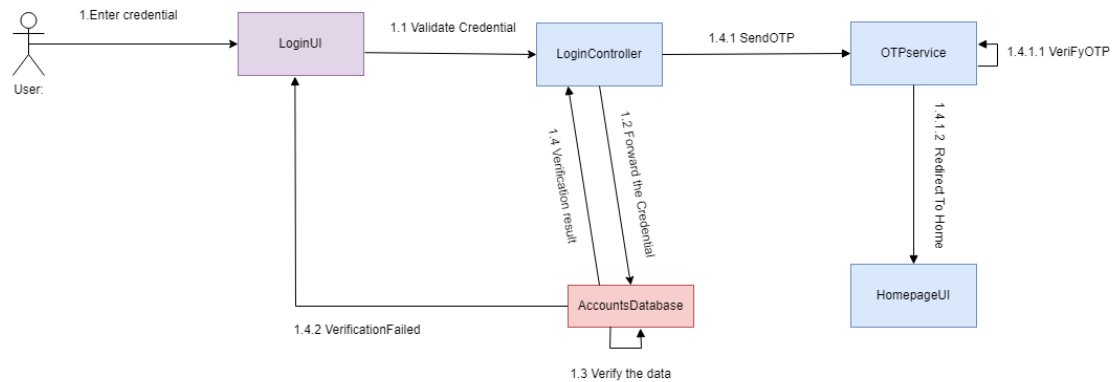


Figure 33:Collaboration for login

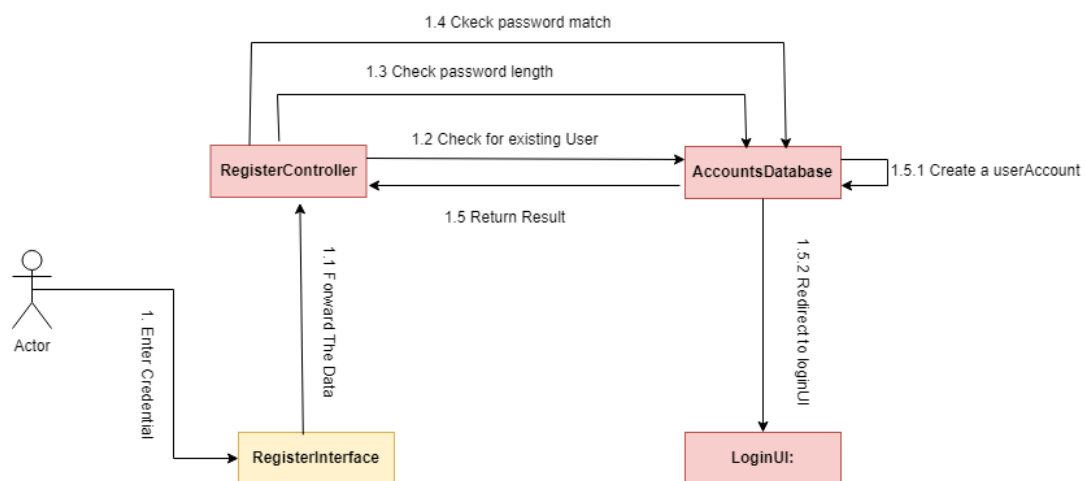


Figure 34:Collaboration for registration

4.9 Class Diagram:

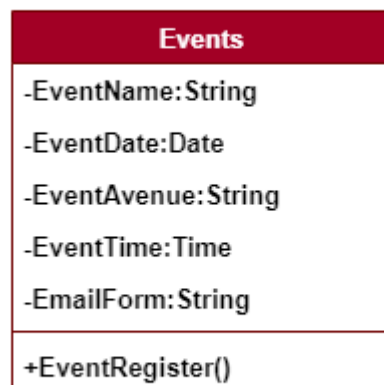


Figure 35:class diagram for Event

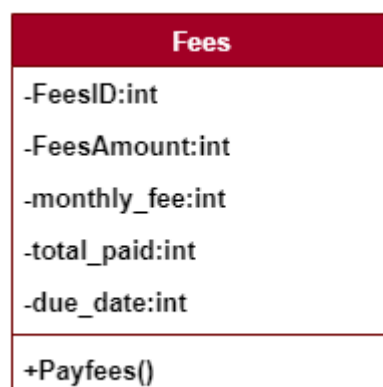


Figure 37:Class Diagram for Fees

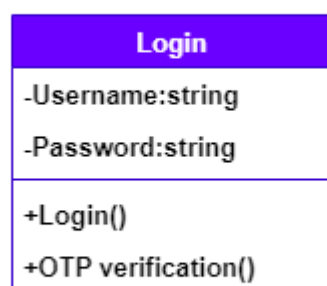


Figure 36:Class Diagram for login

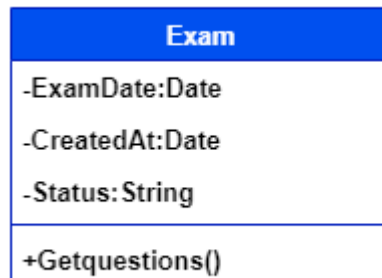


Figure 38:ClassDiagram for Exam

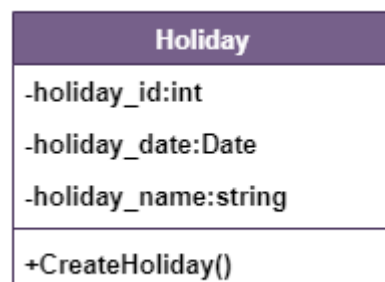


Figure 39: Class Diagram for holiday

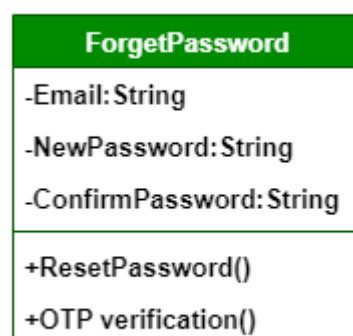


Figure 40:ClassDiagram For forget Password

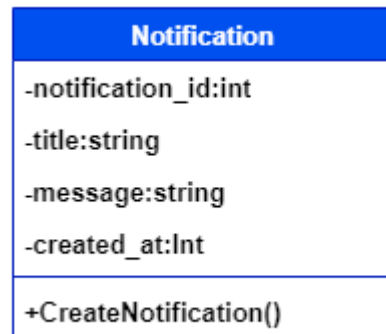


Figure 41:Classdiagram For Notifications

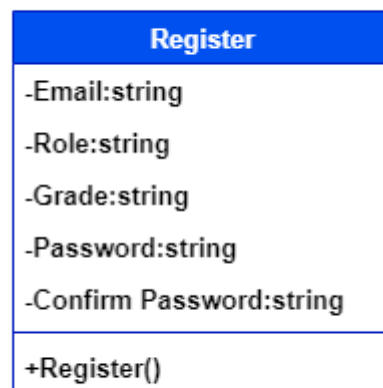


Figure 42:ClassDiagram For Register

4.10 Wireframe:

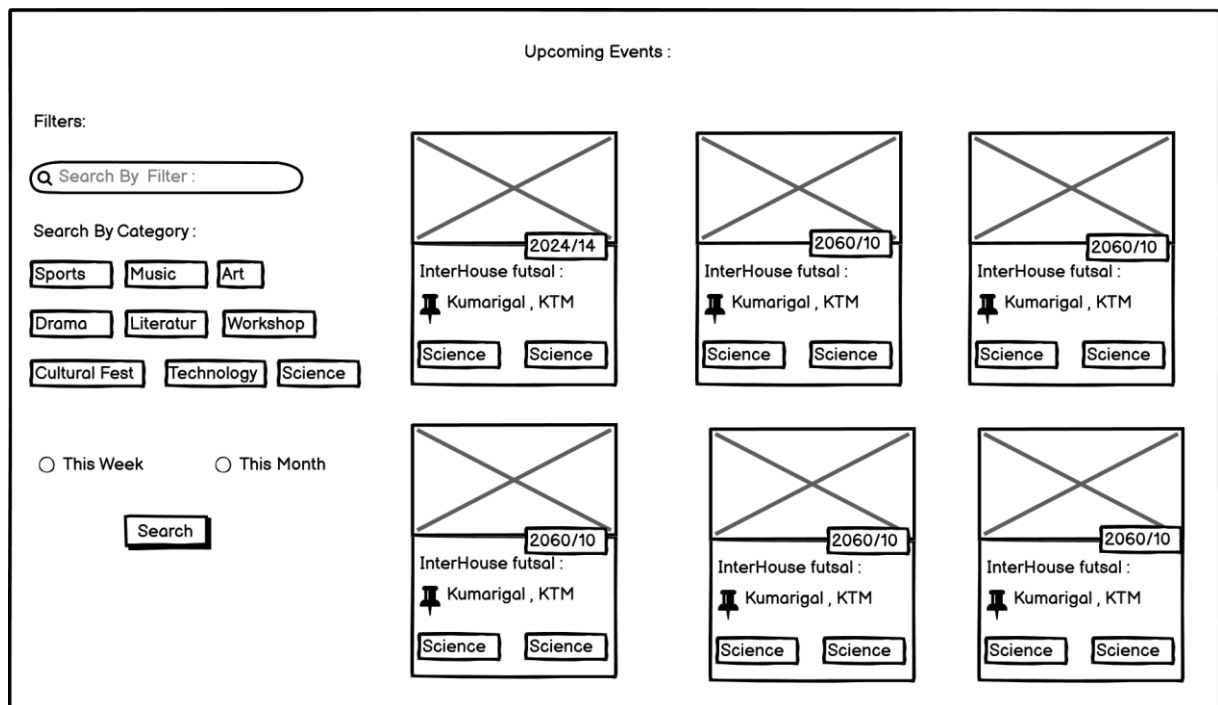


Figure 43: Wireframe for Events page

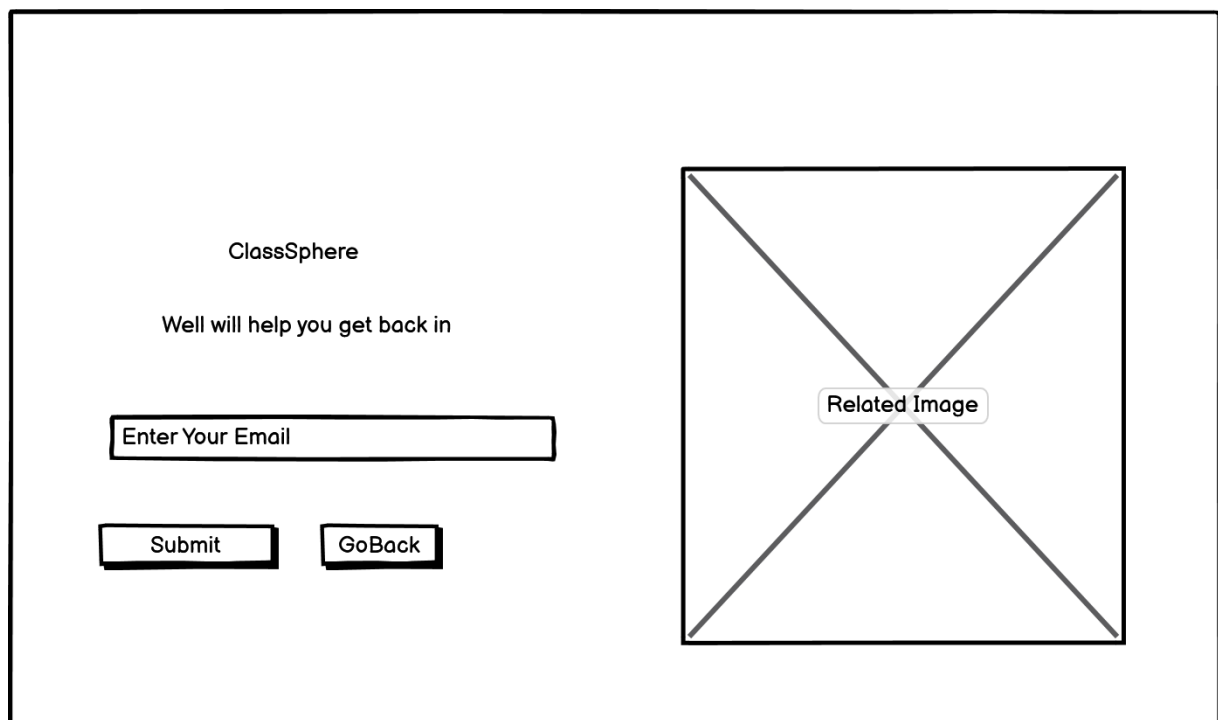


Figure 44: Wireframe for Forgot Password Page

Holiday List:

The below Listed Holiday will be applicable to all the Students ,Staffs and teachers of the organization :

Office Time:
 Summer Time [Sunday-Friday : 9:00 AM - 5:00 AM]
 Winter Time [Sunday Friday : 9:00 AM 3:00 AM]

Break Time :
 Class [1-5] : 12:00 PM
 Class [5-10] : 12:30 PM
 [+2 onward] : 1:00 PM

SN	Occassion	Date	Day
1	New Year	2025-01-01	Wednesday
2	Independence Day	2025-07-04	Friday
3	Christmas	2025-12-25	Thursday
4	Thanksgiving	2025-11-27	Thursday
5	Labor Day	2025-09-01	Monday

Figure 45:Wireframe for Holiday Page

ClassSphere

Login and JoinUs today

UserName

PassWord

[Don't have an account? Sign up now !](#)

Login

ForgotPassword

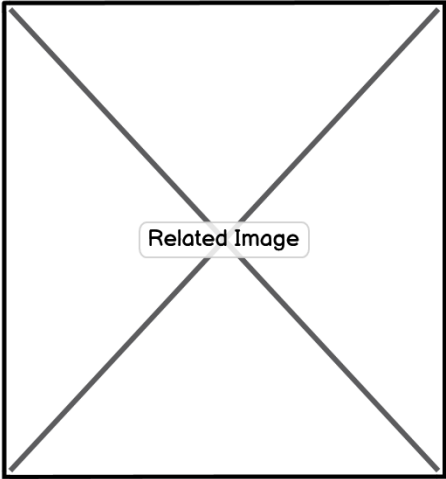
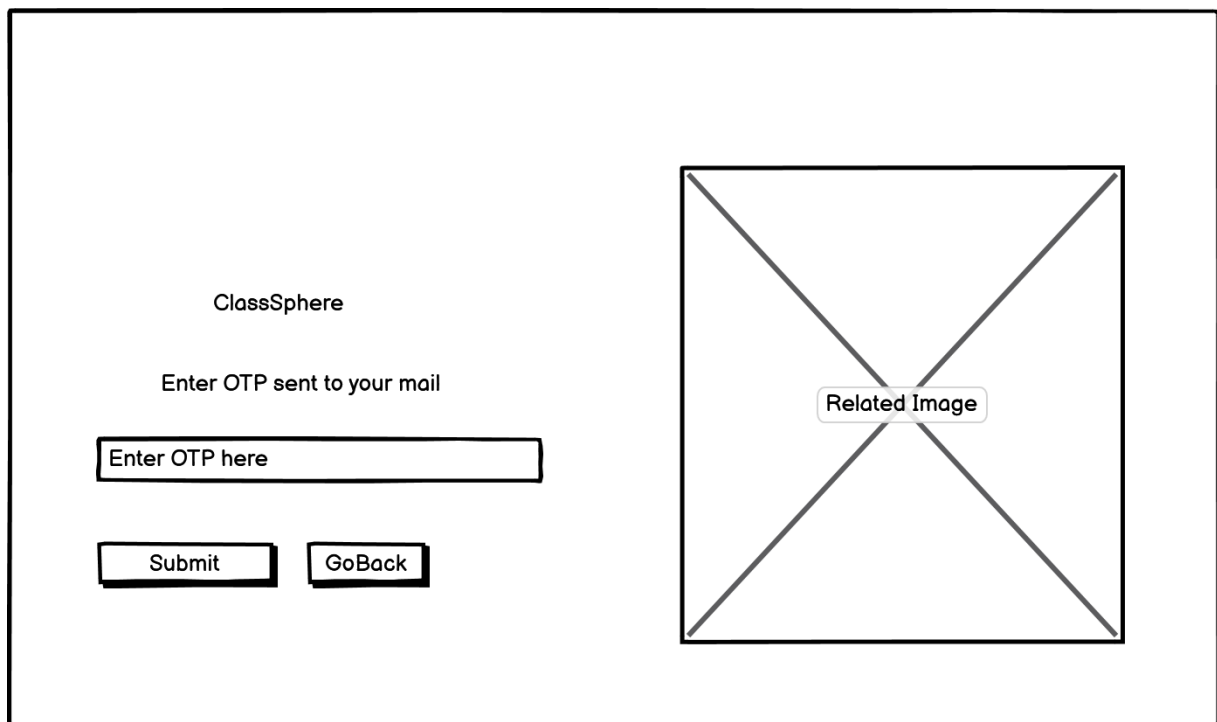


Figure 46:Wireframe for Login



ClassSphere

Enter OTP sent to your mail

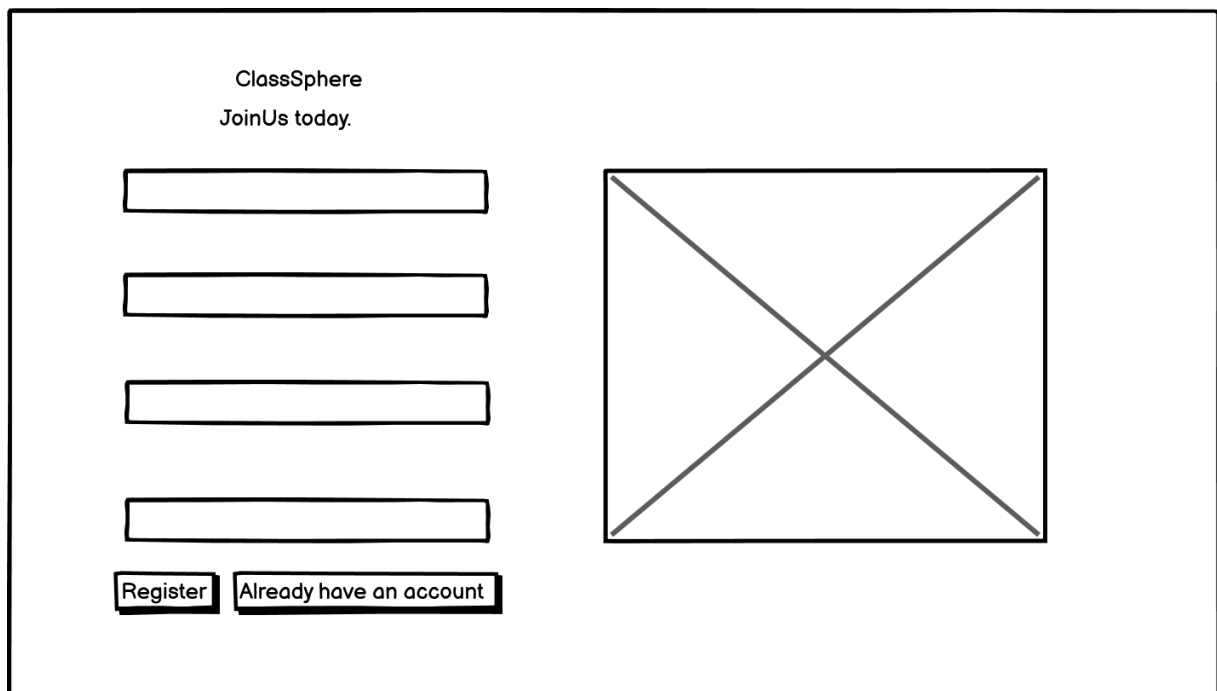
Enter OTP here

Submit GoBack

Related Image

This wireframe represents an OTP verification page. It features a header 'ClassSphere' and a sub-header 'Enter OTP sent to your mail'. Below these is a text input field labeled 'Enter OTP here'. At the bottom left are two buttons: 'Submit' and 'GoBack'. On the right side, there is a large square placeholder for an image, indicated by a diagonal cross and the text 'Related Image'.

Figure 47: Wireframe for OTP page



ClassSphere

JoinUs today.

Register Already have an account

This wireframe represents a registration page. It features a header 'ClassSphere' and a sub-header 'JoinUs today.'. Below these are four stacked text input fields for registration details. At the bottom left are two buttons: 'Register' and 'Already have an account'. On the right side, there is a large square placeholder for an image, indicated by a diagonal cross.

Figure 48: Wireframe for Register

 Basic Information	FullName :	
 Password	UserName:	
 Transactions	Email:	
	This is your primary email address and will be used to send notification emails. Change Email Address	
	Phone:	
	Address :	

Figure 49:Wireframe for profile

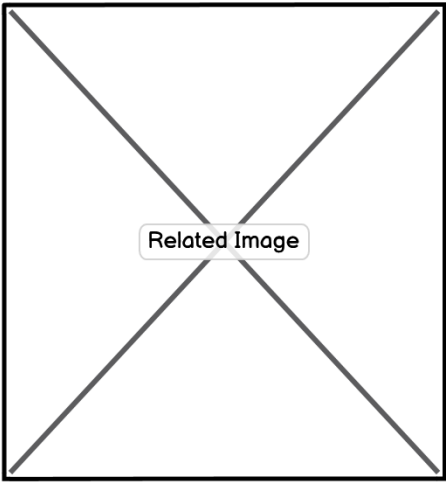
ClassSphere		
Login and JoinUs today		
UserName		
PassWord		
Don't have an account? Sign up now !		
Login	ForgotPassword	

Figure 50:Wireframe for Login

4.11 UI designs:

Login-Frame

Welcome Back

Welcome back! Please login to your account.

Something@gmail.com

☐ Remember Me



Figure 51: UI design for login Page

RegisterFrame

Let's Get Started!

A world of opportunities . Join us today!

SignUp as

Grade

Something@gmail.com

[Already have an account ? Login](#)



Figure 52: UI for SignUp page

OTP-page

Welcome Back!

Enter Your OTP sent in your Mail.

SubmitGo Back



Figure 53: UI for OTP verification Page

Forgot-password

Welcome Back!

We will help you get back on Track.

SubmitGo Back



Figure 54: UI for forget Password page

4.12 Development Status With Proof :

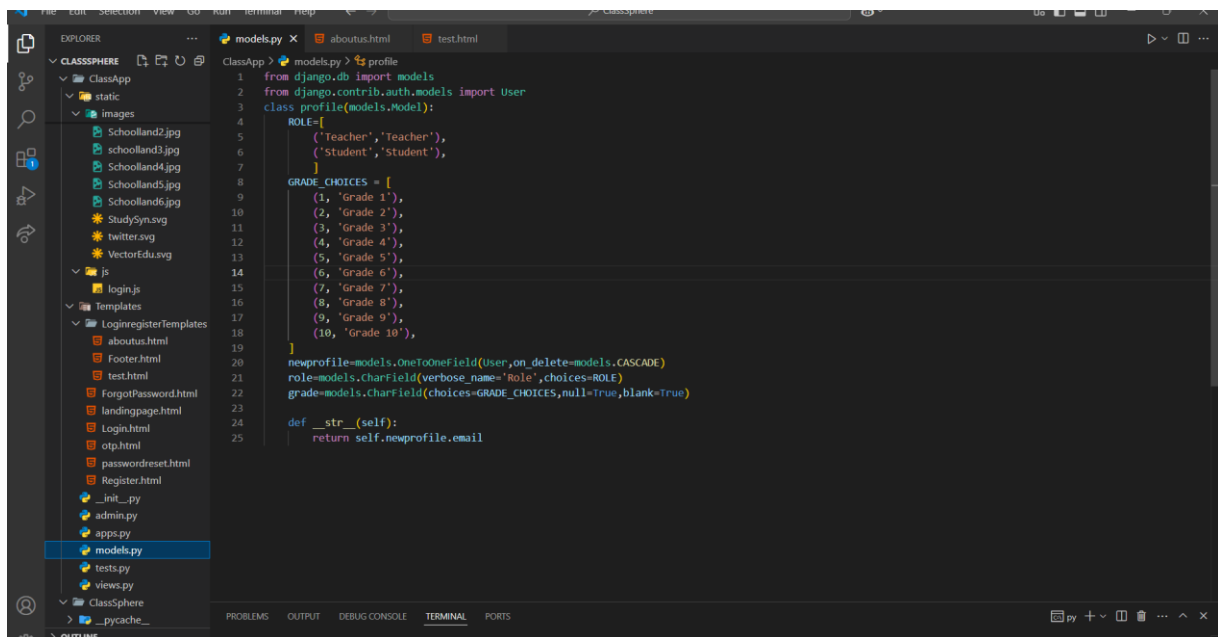


Figure 55: User Extended Model

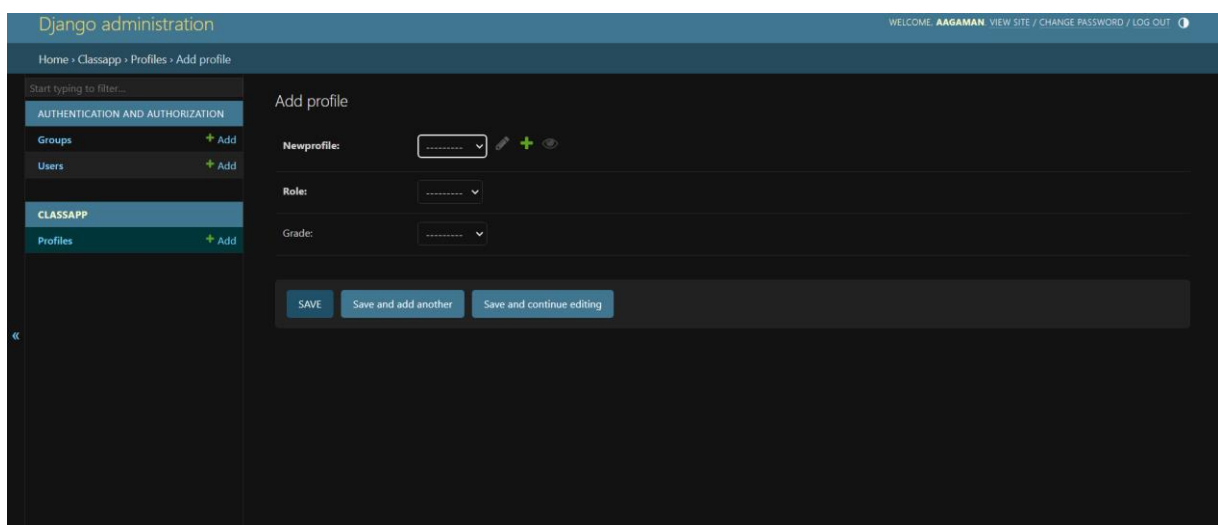


Figure : Django ORM data in Admin page

Let's Get Started!
A world of opportunities . Join us today!

Student

Already have an account? [Login](#)



Figure 56: Password length less than 8

Let's Get Started!
Use a stronger password

Signup as

Already have an account? [Login](#)

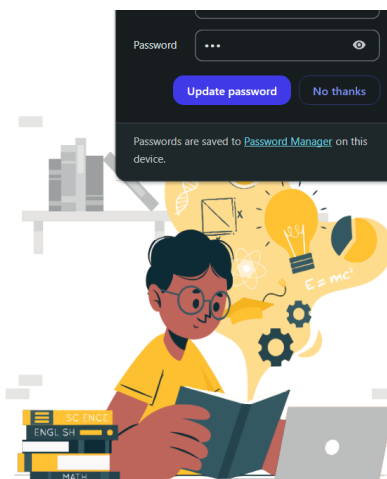
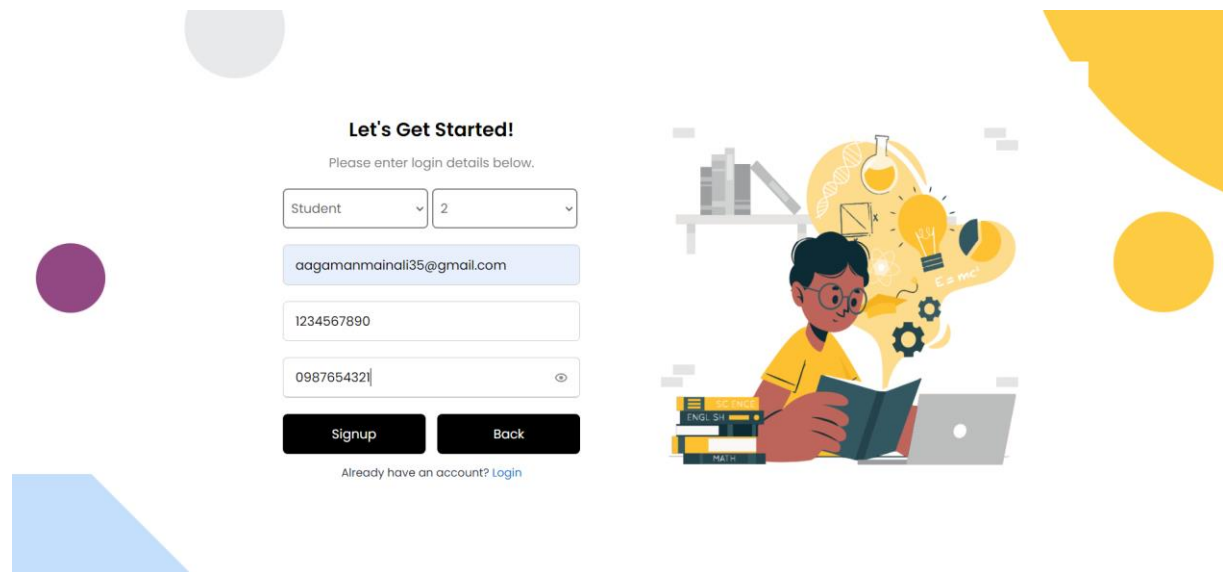


Figure 57: Error message



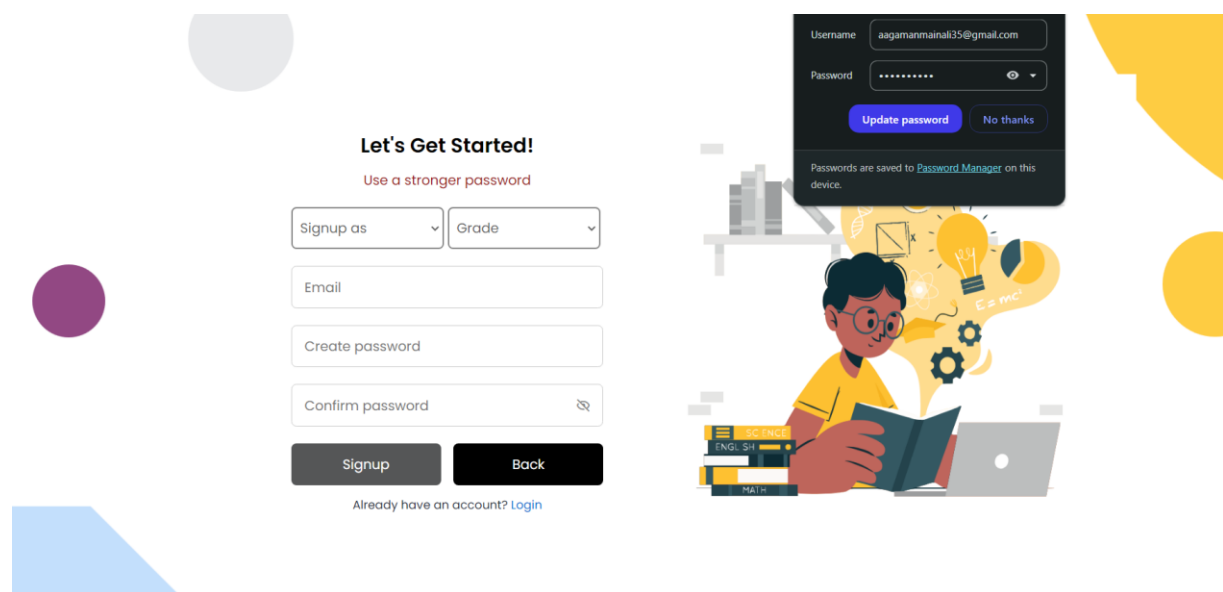
Let's Get Started!
Please enter login details below.

Student

Already have an account? [Login](#)

The login form is displayed on a white background with a yellow and purple geometric design on the right. It includes a title 'Let's Get Started!', a subtitle 'Please enter login details below.', and four input fields: 'Student' (a dropdown menu with '2' selected), 'Email' (containing 'aagamanmainali35@gmail.com'), 'Phone Number' (containing '1234567890'), and 'Password' (containing '0987654321' with an eye icon). Below the inputs are 'Signup' and 'Back' buttons. At the bottom, there is a link 'Already have an account? Login'.

Figure 58: Validation With different Password



Let's Get Started!
Use a stronger password

Signup as

Already have an account? [Login](#)

Username

Password

Passwords are saved to [Password Manager](#) on this device.

The signup form is displayed on a white background with a yellow and purple geometric design on the right. It includes a title 'Let's Get Started!', a subtitle 'Use a stronger password', and four input fields: 'Signup as' (a dropdown menu with 'Grade' selected), 'Email' (containing 'Email'), 'Create password' (containing 'Create password'), and 'Confirm password' (containing 'Confirm password'). Below the inputs are 'Signup' and 'Back' buttons. At the bottom, there is a link 'Already have an account? Login'. A modal window is open over the form, showing a 'Username' field with 'aagamanmainali35@gmail.com', a 'Password' field with '*****', and 'Update password' and 'No thanks' buttons. Below the modal, there is a message 'Passwords are saved to Password Manager on this device.'.

Figure 59: Validation Failed

:



Let's Get Started!

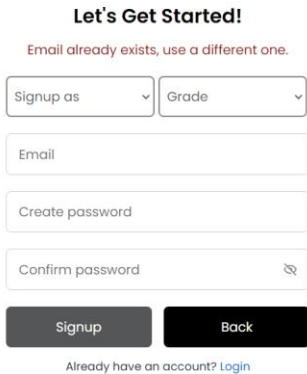
Please enter login details below.

Teacher 1

Already have an account? [Login](#)



Figure 60: Using the user that already exists:



Let's Get Started!

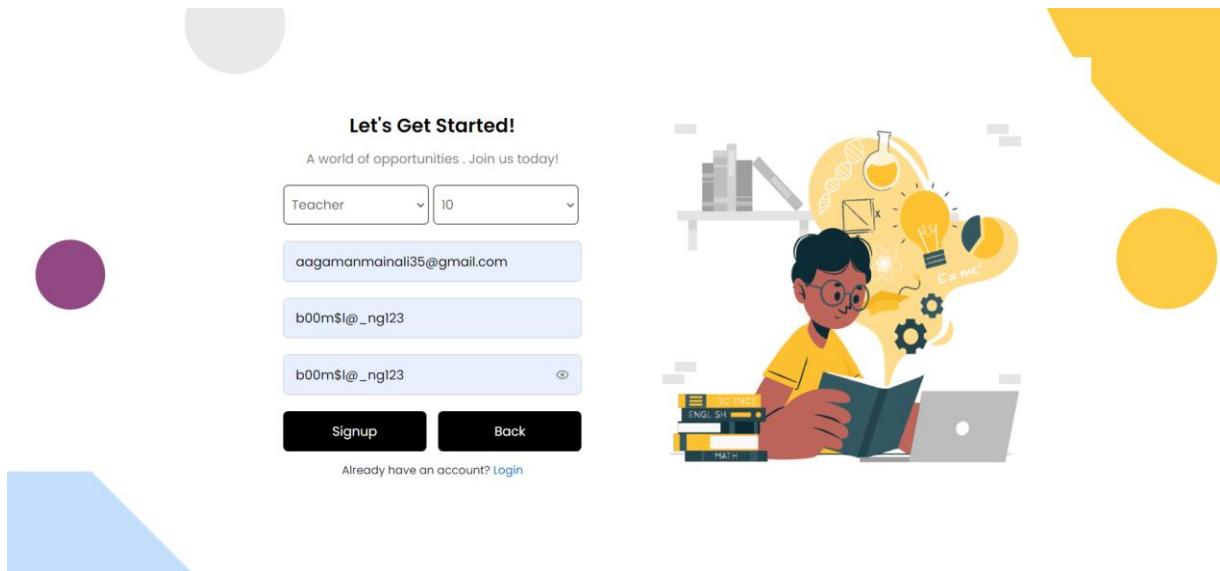
Email already exists, use a different one.

Signup as Grade

Already have an account? [Login](#)



Figure 61: Validation failed due to Unique constraint



Let's Get Started!
A world of opportunities . Join us today!

Teacher

Already have an account? [Login](#)



Figure 62:SignIN with Proper Data

Django administration

WELCOME, AAGAMAN VIEW SITE / CHANGE PASSWORD / LOG OUT

Home > Authentication and Authorization > Users

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

- Groups [+ Add](#)
- Users** [+ Add](#)

CLASSAPP

- Profiles [+ Add](#)

Select user to change

Action: Go 0 of 2 selected

	USERNAME	EMAIL ADDRESS	FIRST NAME	LAST NAME	STAFF STATUS
☐	aagaman	-	-	-	☒
☐	aagamanmainali35	aagamanmainali35@gmail.com	-	-	☐

2 users

FILTER

- Show counts
- By staff status
 - All
 - Yes
 - No
- By superuser status
 - All
 - Yes
 - No
- By active
 - All
 - Yes
 - No

[ADD USER +](#)

Figure 63:New user created

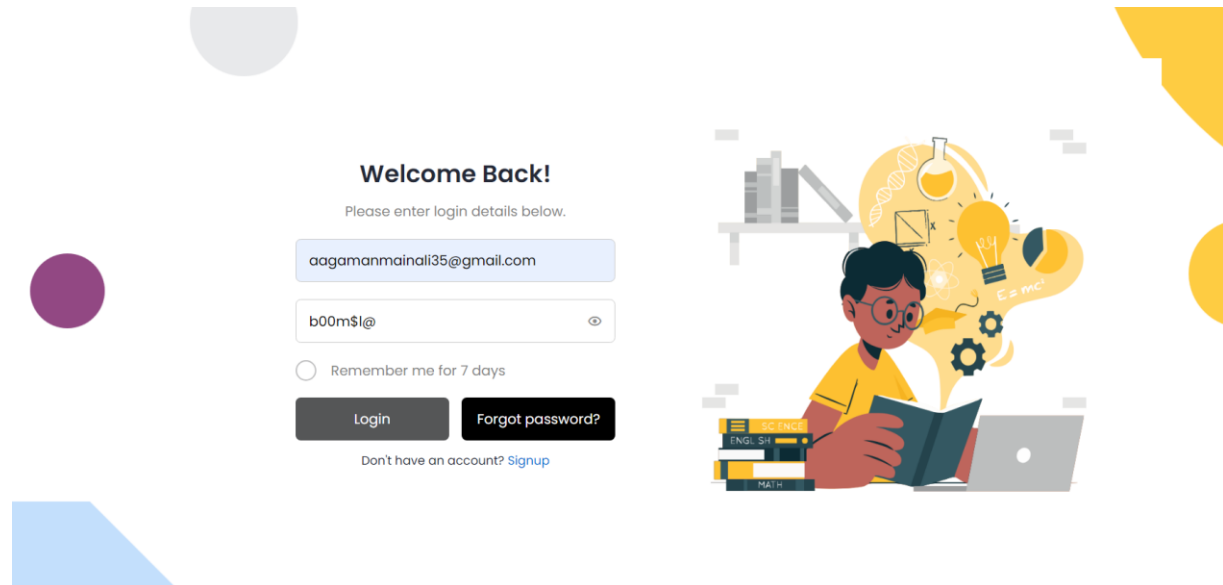


Figure 64: Using Wrong Password

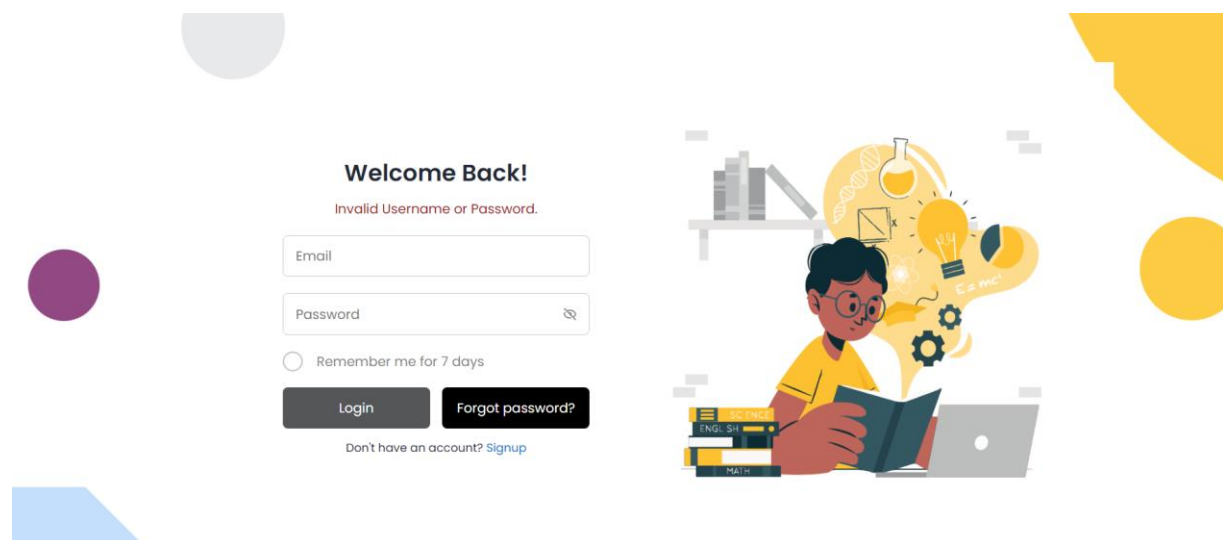


Figure 65: Login Validation Failed

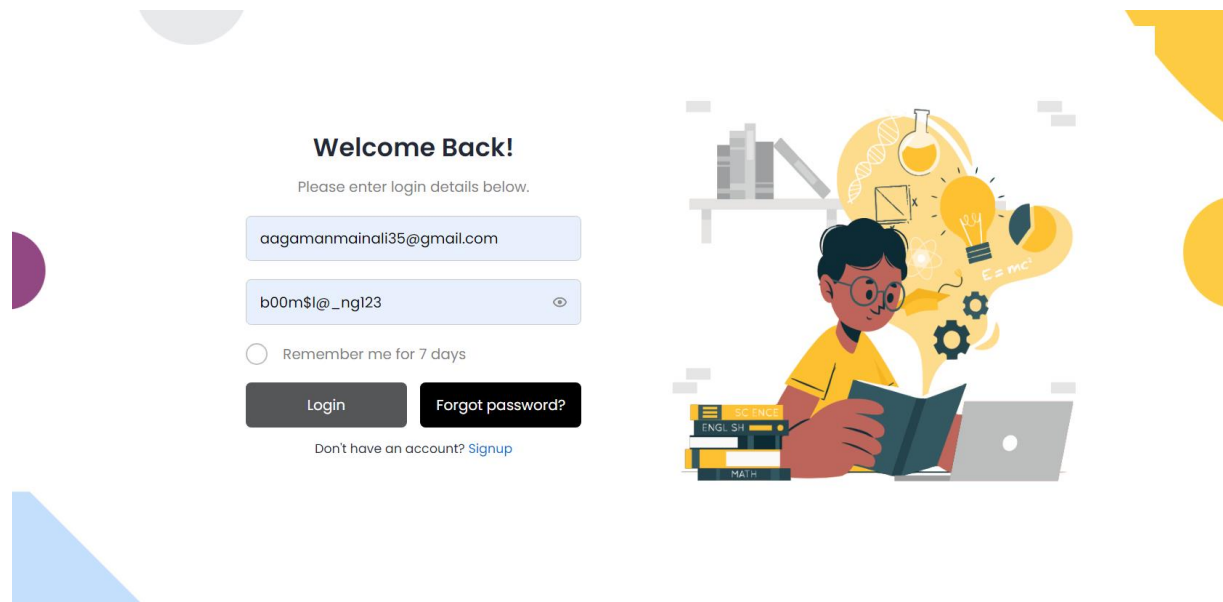


Figure 66:Using Correct Password field

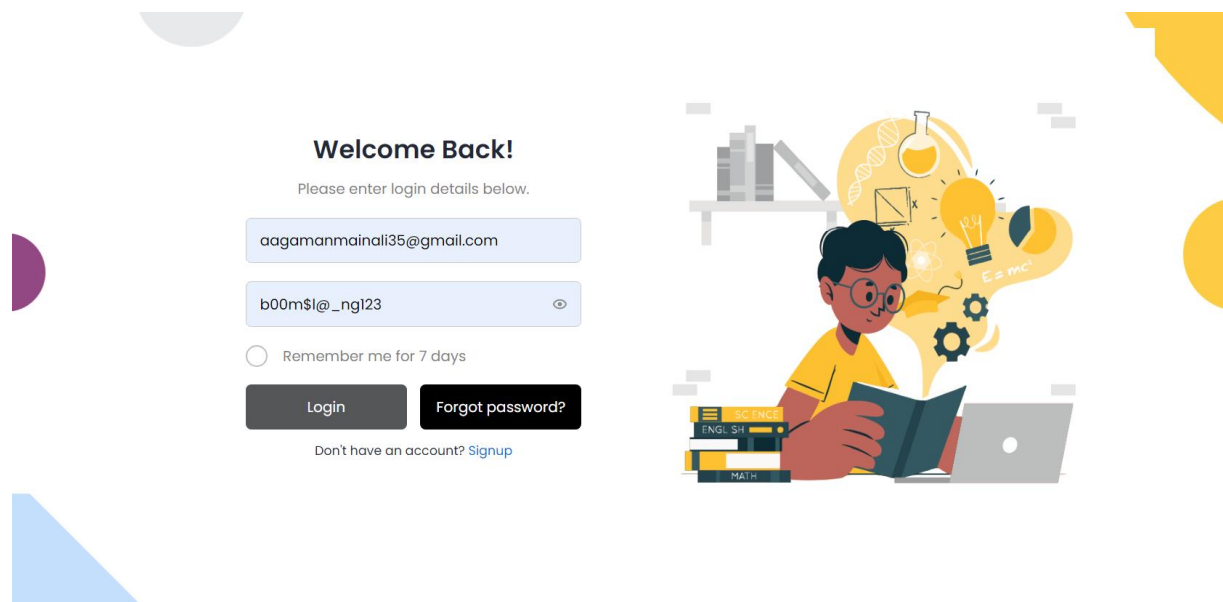


Figure 67:Redirected To login

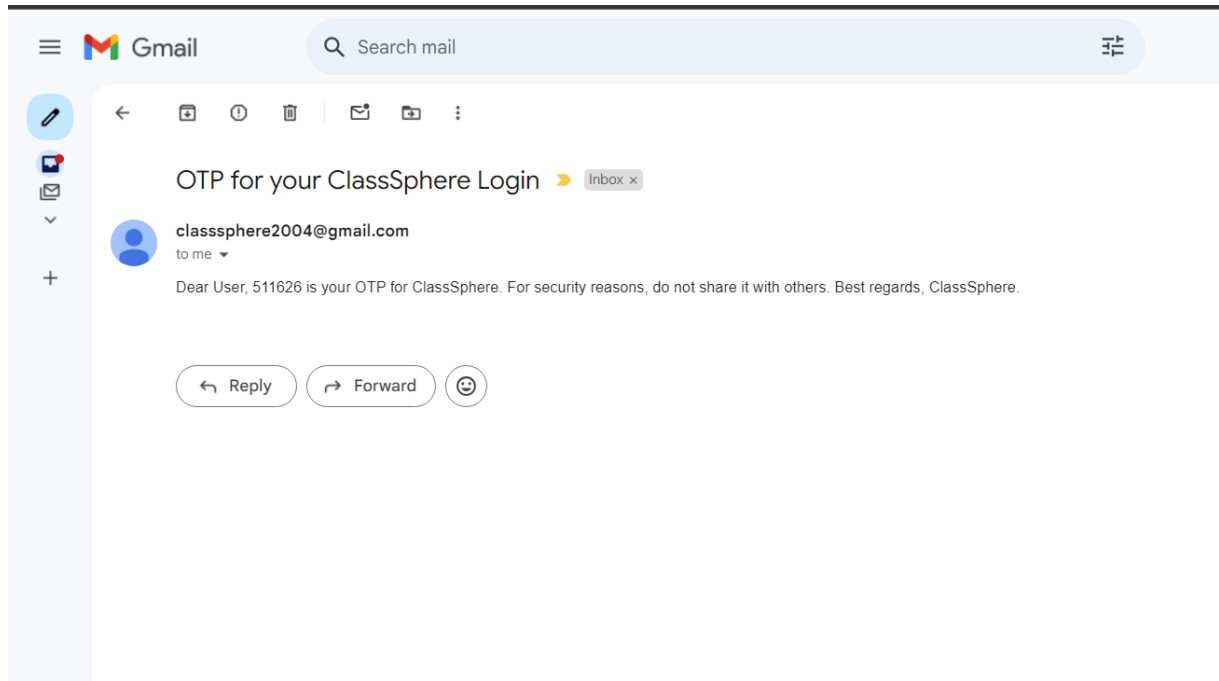


Figure 68:OTP sent

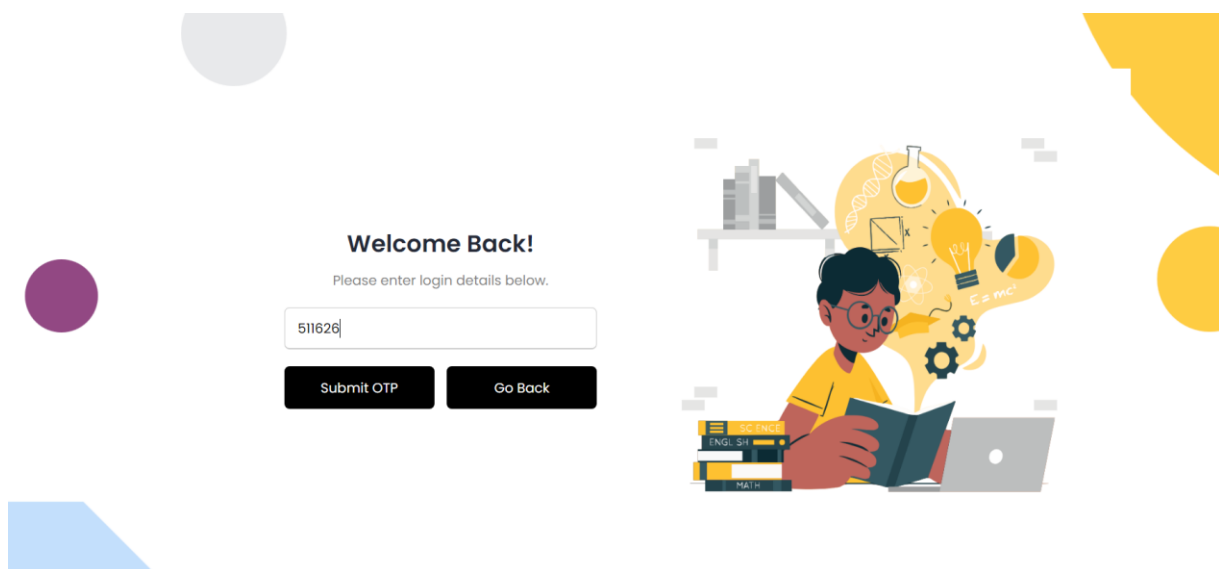


Figure 69:Entered OTP

ClassSphere

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"StudySync transformed our family's learning journey. Safe, interactive, and progress tracking features are invaluable for parents."

"StudySync, a true game-changer! Flexible content, tailored assessments, and excellent support. Integral to our school district's success."

Figure 70:Redirected to login Page

5.Future Works:

5.1 Remaining task analysis with Gantt Chart:

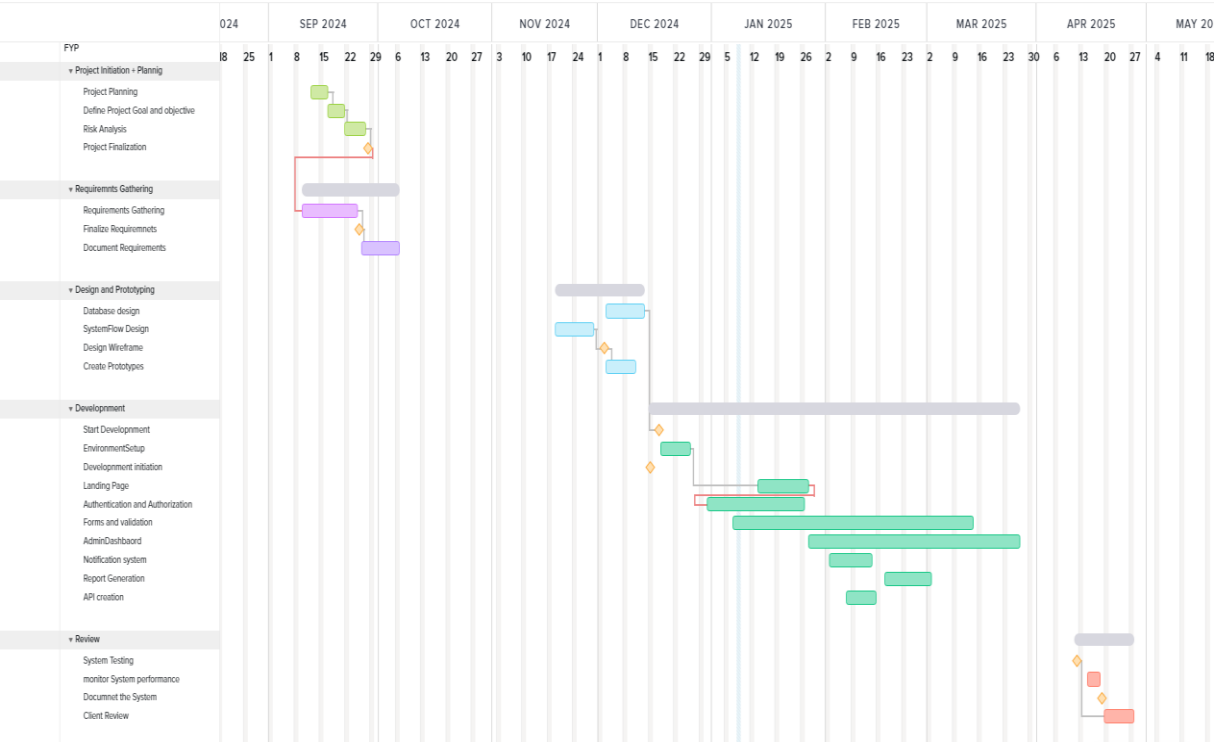


Figure 71:Updated Gantt chart

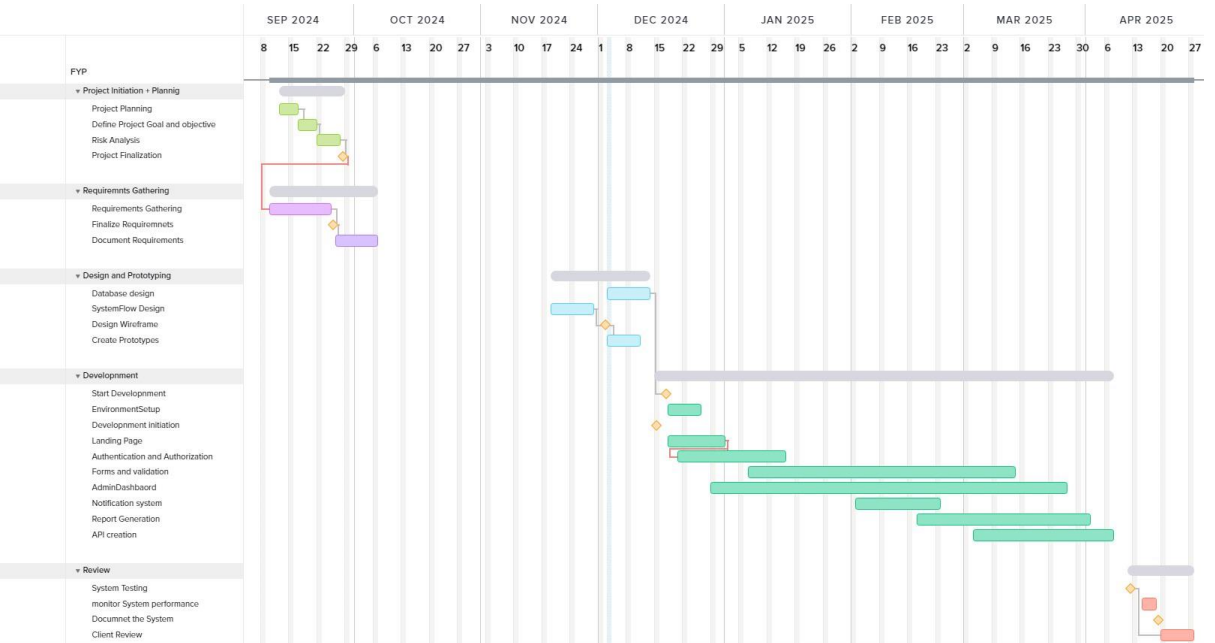


Figure 72: Old gantt chart

Analysis of the Gantt charts:

In the project proposal phase the first initial Gantt chart was produced, but later was changed to much more accurate and efficient Gantt chart to be used in the project. The changes were made due to time constraints identified in the previous schedule: certain aspects of the development were not given enough time.

The new version of the Gantt chart includes a separate testing block and more time for coding and improving the system. All these added improvements are focused on improving the general valuation of the project as well as increasing the prospects of delivering the project on the best manner possible. Though it goes a little off the mark in terms of the length of the project, the revised timeline significantly minimizes the two flaws of having an ill-defined and loose structure of the project into small segments which may harm the overall flow of the project or concentrating on too few large milestones at the expense of losing sight of the overall objectives of the project.

5.2 Phase To complete:**1. Event Management System:**

Develop an easy to navigate design of the Mock Event Page that allows people to view the events and create/mock events.

- ❖ Backend Integration: Create and edit events, delete while being able to delete them.
- ❖ Notifications: It is also necessary to make notifications regarding new or upcoming events for app users.

Testing: Perform tests to check if events will be displayed, updated and managed in future as expected by the system designers.

2. Notification System:

- ❖ Development: Integrated a notification system in order to notify users, reminders, updates, events, or impending deadlines both through their email and within the application.
- ❖ Customization: Improve the notification defaults for enabling different alerts, which may be selected or deselected according to interest.
- ❖ Testing: Certify the notification delivery mechanism before sending alerts to avoid sending redundancy or delayed notifications.

3. Report Generation:

- ❖ Automated Reports: Create a reporting framework that will enable the administrators to create ad hoc reports on attendance, events and any other statistic.
- ❖ Export Options: Include download files templates that will allow a user to download the current report in different formats such as PDF, Excel among others.
- ❖ Testing: Make sure they are correct in content, stylized as needed and mobile optimized.

4. Admin Dashboard:

- ❖ Development: Develop an effective admin dashboard to enable the administrators to go about the business of adding users, events and managing system settings.
- ❖ Role-Based Access Control: Set up proper restrictions of access in order to later control access to materials containing classified data.
- ❖ Testing: Do not forget to test all of the admin options in order to avoid any violations of data integrity and miscellaneous issues.

5.Examination System

- ❖ Develop a user-friendly interface for students to take exams and view their results.
- ❖ Provide separate interfaces for administrators/teachers to create, update, and manage exams.
- ❖ Implement role-based access to ensure only authorized personnel can manage exams.
- ❖ Enable features for creating exams with multiple question formats such as multiple-choice, short-answer, and essay-type questions.

6.FeesPayment:

- ❖ The other important feature of the system should be to enable users to easily access any pending fees and existing payment history as well as fee due dates.
- ❖ Provide administrators with tools to configure fee structures, apply scholarships, discounts, or financial aid, and log offline payments.

7. Notification

- ❖ Develop a notification system to send alerts, reminders, and updates about events, deadlines, and important activities via email and in-app notifications. Allow users to customize notification preferences to enable or disable specific alerts based on their interests. Ensure timely and accurate delivery of notifications by thoroughly testing the system to avoid delays or redundancies.

8. Holiday Creation

- ❖ Emailing, sending push notifications to users when it is close to the holiday or the changes in working hours.
- ❖ Offer users to see the information about the holiday and settings of receiving notifications concerning this or that holiday.

5.3 Detailed and Coherent plan of work:

Phase 1: Planning & Design (Duration: 1 Month)

Duration: January 8, 2025 – February 8, 2025

Task to achieve: Define requirements and objectives for the holiday notification system. Create detailed user stories, flowcharts, and wireframes for holiday notifications.

Identify key holidays and working hour changes to notify users about.

Research and finalize the email and push notification delivery mechanisms.

Phase 2: Development of Notification System (Duration: 1 Month)

Duration: February 9, 2025 – March 9, 2025

Task to achieve: Implement the backend system to schedule and send holiday notifications. Develop user interface for displaying holiday information.

Integrate email and push notification services to deliver alerts about holidays or working hour changes.

Develop settings page for users to customize their notification preferences regarding specific holidays

Phase 3: Customization & User Preferences (Duration: 1 Month)

Duration: March 10, 2025 – April 9, 2025

Task to achieve: Develop functionality to allow users to view details of upcoming holidays. Test the delivery of notifications and ensure they match user preferences.

Phase 4: Testing, Optimization & Deployment (Duration: 20 Days)

Duration: April 10, 2025 – April 30, 2025

- Perform thorough system testing for both email and push notifications.
- Verify accuracy of holiday information and notification delivery.
- Conduct testing to ensure system stability under different user loads.
- monitor performance and address any issues.

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7. Appendix:

7.1 Client System requirement Letter:

Client Agreement Letter :

Date: November 21, 2024

To,
Aagaman Mainali

Subject: Letter of Confirmation

Dear Aagaman,

I am writing to formally confirm my agreement for the client role in your project. The project requirements have been discussed in detail and documented as per my specifications to ensure alignment with the intended objectives.

I look forward to collaborating with you on this project and am confident in your ability to deliver a high-quality solution. Please do not hesitate to contact me should you require any further clarification or input during the course of the project.

Requirements list:

Functions	Description	Notes
1. Login/ register (captcha or OTTP)	-Users will log in by receiving an OTP (One-Time Password) on their registered mobile number or email. This will enhance security and avoid the need for remembering passwords. - New users must complete a captcha challenge to prove they are human during the registration process. This prevents bots from creating fake accounts.	
2. Enrollment form	- Users will fill out an online form for enrollment, such as a registration form for Admission. The submitted data will then be compiled into a PDF format for the user to download and fill form manually or use google form for online form .	# Use google Forms For online entry # downloadable PDF To fill form manually

Figure 73:Client Letter page one

3. Events with registration and news page	- The system will maintain a list of upcoming events displayed on the website for users to view. - Users can register for upcoming events using online forms. - A news section will display updates or articles related to the system or upcoming events.	# one Student can take part in multiple events # News section with upcoming events holidays and programs
4. Examination System	- Small End Exams with Multiple Choices: The system will allow (students) to take exams. The exams will be objective type (multiple choice questions), and their responses will be recorded and processed automatically.	# Online tests for MCQ
5. Attendance sheets (Extract from excel)	- Extract Data from Excel: Attendance data will be imported from Excel sheets, and the system will automatically generate individual attendance reports for each class or course. - Create Attendance Sheets for Classes: Based on the imported data, the system will create structured attendance sheets that can be viewed or downloaded by instructors or students.	# From a excel file containing attendance extract it and provide attendance based on class and student
6. User profile Completion	- Profile Update: After registering, users will be able to update their profile details such as personal information, contact details, etc. - Profile Completion After Registration: Users will be required to complete their profiles by adding essential information after registration.	
7. Fees Collection	- Fee Collection Panel: The system will allow administrators to manage and track fees. It will include functionalities to create fee structures, set deadlines, and generate invoices. - User Fee Management: Students can access their fee details, including pending and paid amounts. Payment history will be available to the users.	# manual fee collection not online e-sewadoesn't support

Figure 74: Client Letter page Two

8. Notification system	- Notification System for Events: The system will send notifications to users for upcoming events, changes, or deadlines. Notifications could be sent via email, SMS, or push notifications on the website or app.	
9. Admin page With CMS features	- Create Holiday: Administrators can set holidays or non-working days. - Register an Event: Admins can register new events and make them available for user registration. - Remove Passed Out Users: Admins can set a time period after which users (like who have completed their course are automatically removed from the system. - Create New Notification: Admins can create new notifications (e.g., for event reminders or important updates) and send them to all relevant users. - Create Attendance Data and Reports: Admins will have access to attendance data and can generate reports on attendance for specific courses or events.	# Admin page with all content management system for the whole websites

Client Details:Name: *Ujjal Lamichhane*Address: *Makalbari, Kathmandu*Phone: *982594 1070*Email: *ujjallamichhane@gmail.com*Signature: *[Signature]*

Figure 75: Client Letter page Three

7.2 Data Dictionary:

7.2.1 User Table

Attributes	Datatype	Constraints
UserID	int	PrimaryKey
UserName	Char(100)	unique
Email	Char(100)	Not nul
DOB	Date	Not null
Address	Char(100)	Not null
Gender	Char(100)	Not null
ContactNumber	Number(10)	Not null

7.2.2 Event Table:

Attributes	Datatype	Constraints
EventID	int	Primary key
EventName	Char(100)	Not null
date	Date	Not null
venue	Char(100)	Not null
Event_form	Char(100)	Venue null

7.2.3 Class Table:

Attribute	Data Type	Constraints
ClassID	Int	Pk
ClassName	Char(100)	Not null

7.2.4 Fees Table:

Attribute	Datatype	Constraints
ClassID	int	FK
UserID	int	FK
FeeID	int	FK

7.2.5 Holiday table:

Attribute	Datatype	Constraints
HolidayID	int	PK
HolidayName	Char(100)	Not null

7.2.6 FeeStructure:

Attribute	Datatype	Constraints
FeeID	Int	PK
FeeAmount	Int	NOT NULL
TotalPayable	Int	NOTNULL
TotalMonthlyPayable	Int	NOT NULL

7.3 Use case Description:

7.3.1 Use Case: Login

Actor(s): User

Description:

Registered users can securely access the platform by logging in with their registered mobile number or email address. The authentication process employs a One-Time Password (OTP) mechanism, sent to the user via SMS, email, or authenticator apps, ensuring secure and password less entry. Users are required to enter the received OTP within a specified timeframe to validate their identity. Successful verification grants access to the platform's features, including browsing content, accessing personalized data, and performing specific actions, such as submitting requests or initiating transactions.

7.3.2 Use Case: Register:

Actor(s): Users

Description:

New people can sign up on the platform and also input some details such as their working email or phone number and the password. The registration info makes account creation more secure through the use of password validation techniques. To stop users typing in the wrong password users are asked to type it and confirm it. The system validates the password based on the following criteria:

❖ Minimum Length Requirement:

For adequate complexity and security, the password should be at least 8 characters in length.

❖ Password Match Validation:

The system ensures that while registering the password entered by the user matches the confirm password field before proceeding to the registration process.

❖ Character Requirements:

The password should use both upper and lower cases, numbers as well as special characters such as (! @, #, \$,&;) for strength not easiness to crack.

7.3.3 Use Case: Fee Payment:

Actor(s): Users

Description:

This use case illustrates a fee payment process by the admin manual entry of payments received in cash or through any method that is not electronic. Users then go to the admin section where they meet all the necessary payments that reflect their respective fee structure depending on whether it includes tuition fees, examination fees or even the other extras of their respective courses.

The admin logs in the system and get the fee details of the user and check the due amount and receive the payment and update the user account. On submission of transaction details, the system changes the payment status and produces a receipt in the form of a receipt that the user can either download or print.

7.3.4 Event Participation

Actor(s): Users

Description:

This use case illustrates a procedure by which a user can participate in multiple events that are hosted by the admins. The admin first create an event instance

7.3.5 Event Participation

Actor(s): Users, Admin

Description:

This use case captures how the users can get involved in multiple events proffered by the admins. Users initially develop an event instance with such details as the name of the event, the description of the event, date, time, and place. The users can then look at these events from the platform and can opt to turn up for the event by having to register for the event.

7.3.6 Use Case: Holiday Creation

Actor(s): Admin

Description:

This use case lays down the procedure through which an admin will be able to create holidays for the institution. The exercise begins with the admin logging into the application's "holiday management" where he or she is required to feed the system with information like, the name of the holiday, the description, date, and importantly whether it is a one-off event, or are occurring event like national holidays every year or on such occasions.

Any time the holiday details are keyed in, the system checks the correctness of the inserted values and implements them into the institution's holiday list. All the students, teachers as well the staff are given notification of the holiday which is going to come.

7.4 Survey Form Responses:

Does it help you Paying you Institutions fees through system or paying it in person ?

20 responses

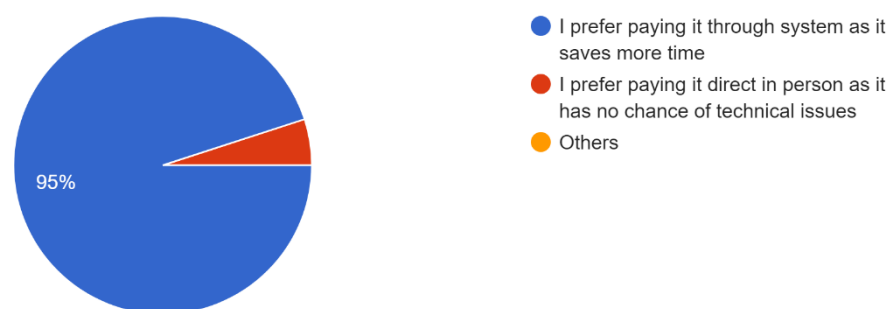


Figure 76:Question 1

Has or Had your school or institution used any management software before?

20 responses

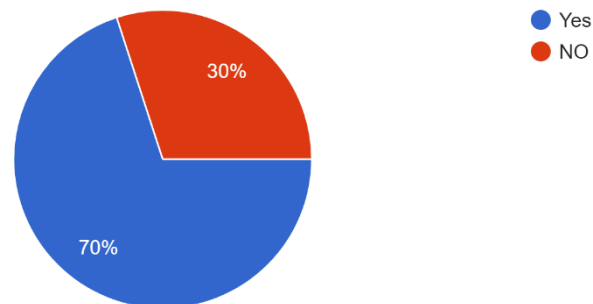


Figure 77:Response 2

Does it help you Paying you Institutions fees through system or paying it in person ?

20 responses

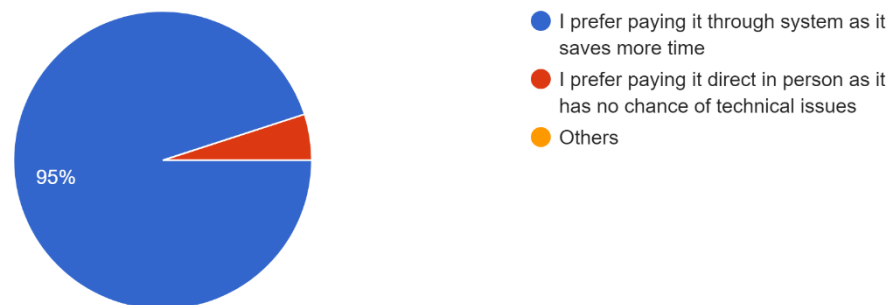


Figure 78:Response 3

How much time do you usually spend on such systems daily?

20 responses

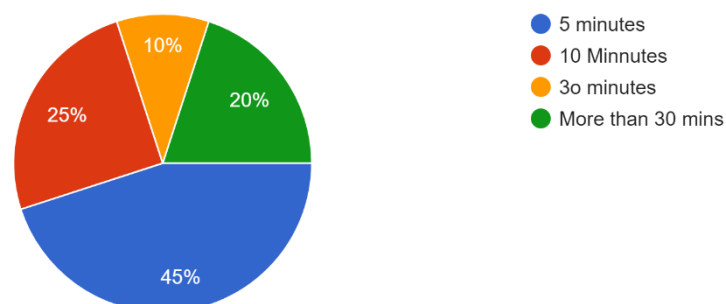


Figure 79:response 4

How satisfied were you with the system?

20 responses

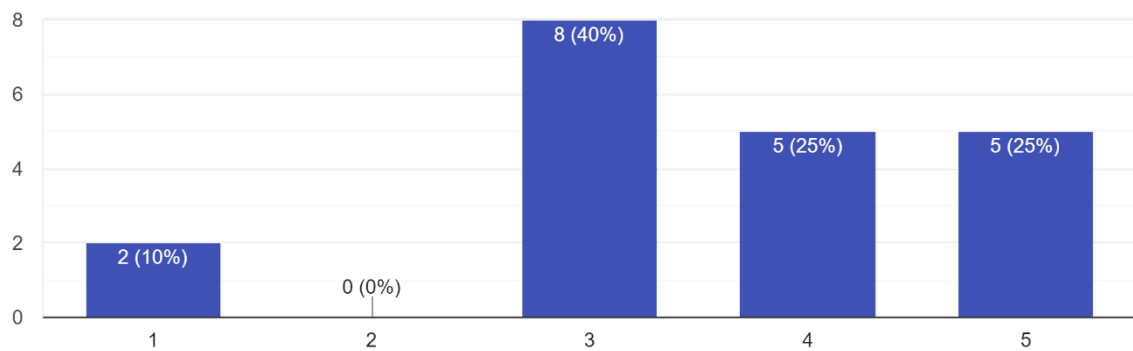


Figure 80:Reponse 5

If you had to choose, would you prefer Management systems or the traditional management method ?

20 responses

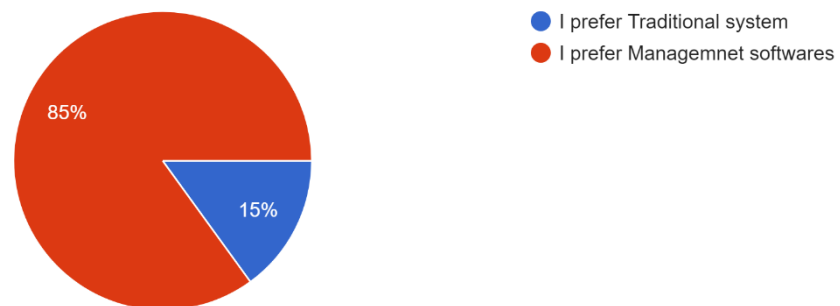


Figure 81:response 6

Did the previous software make managing tasks easier or harder compared to manual processes?

20 responses

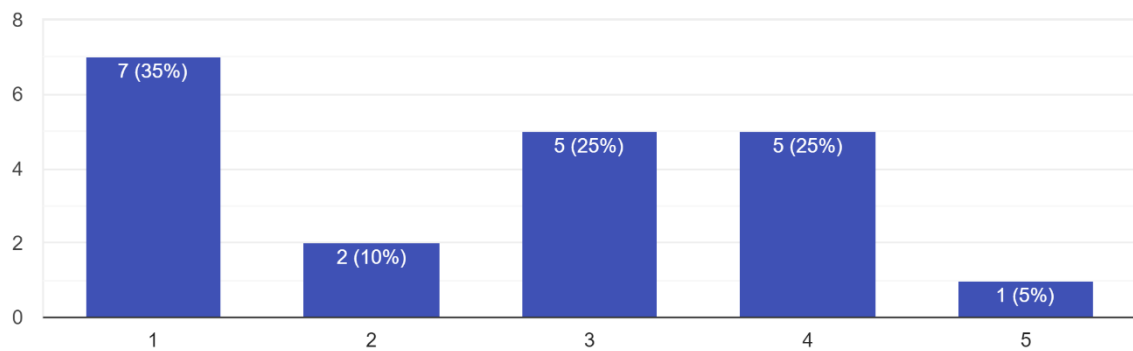


Figure 82:Response 7

What challenges did you face with the previous system?

20 responses

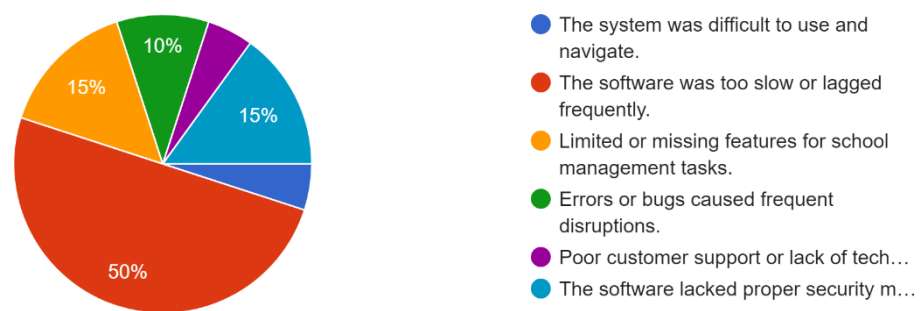


Figure 83:Response 8

Are you familiar with online fee payment systems?

19 responses

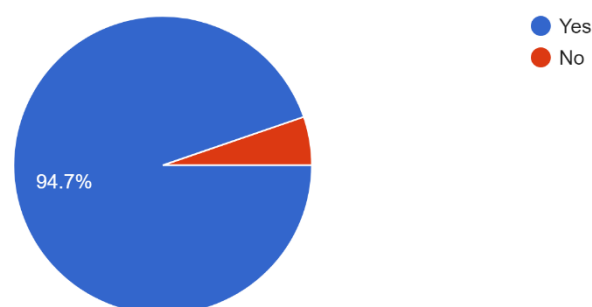


Figure 84:Response 9

Do you prefer to receive timely reminders for pending fees?

19 responses

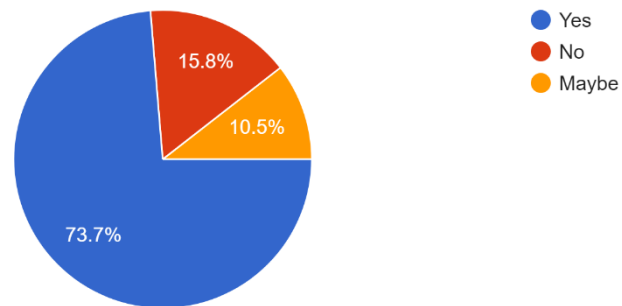


Figure 85:response 10

Do you feel the Two factor Authentication and OTTP verification process is secure?

19 responses

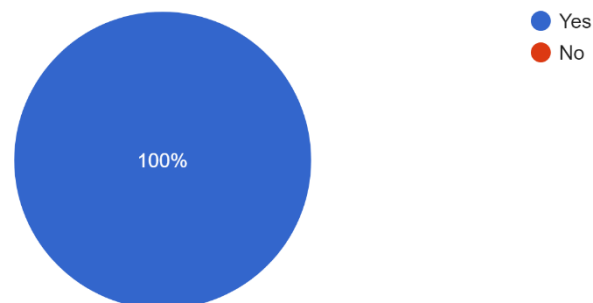


Figure 86:response 11

Do you feel the Two factor Authentication and OTTP verification process is secure?

19 responses

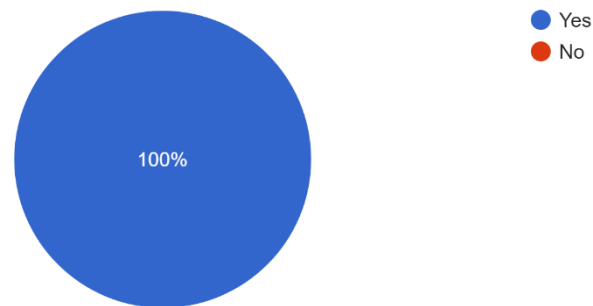


Figure 87:response 12

Would you prefer system sending attendance reports to you

19 responses

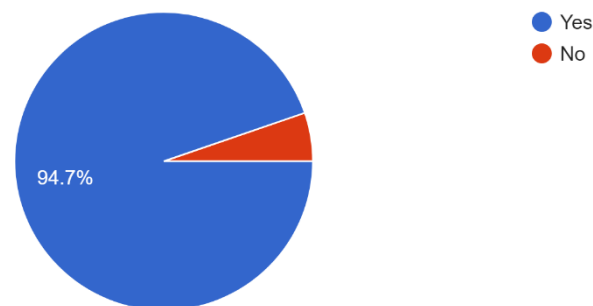


Figure 88:response 13

Is it helpful to have downloadable attendance reports?

19 responses

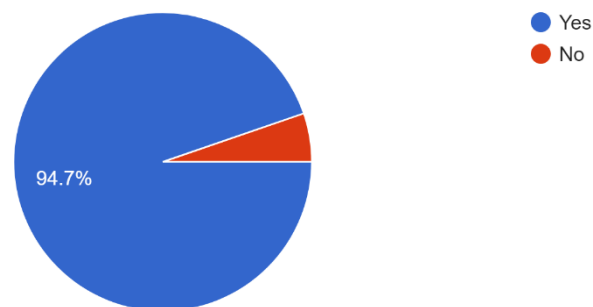


Figure 89:response 14

Do you feel your personal information is safe while using the system?

19 responses

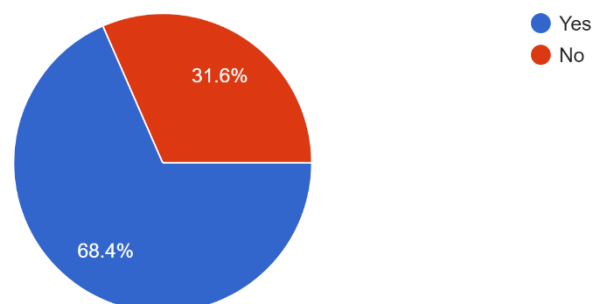


Figure 90::response 15

How often do you participate in events

19 responses

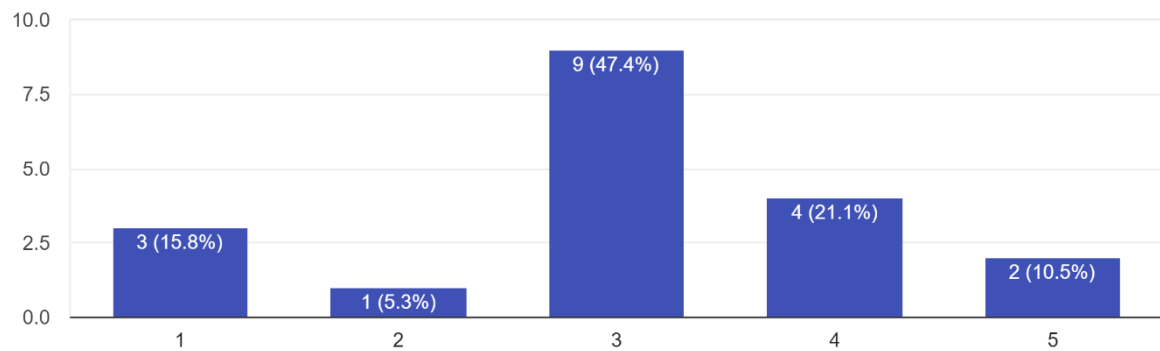


Figure 91:response 16

Do you prefer the process for registering for events through manually or through website

19 responses

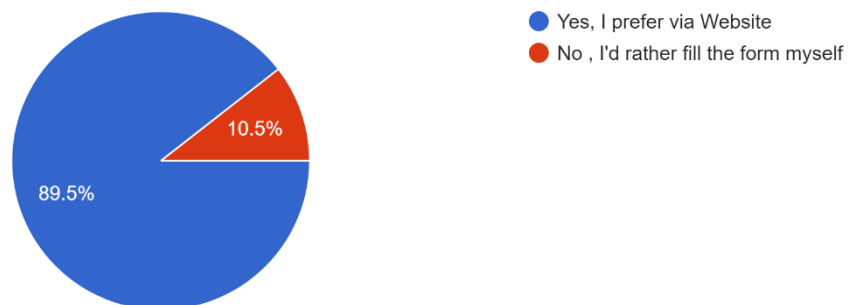


Figure 92:Response 17